

INTRODUCTION TO ICT

Meaning of Information

Information refers to processed or organized data that is meaningful and useful to the person receiving it. It provides context, relevance, and purpose, allowing people to make decisions, solve problems, or understand situations better.

Examples Of Information Kept In A School

- Student Records – Names, ages, attendance, grades, and medical information.
- Teacher Information – Qualifications, timetables, subjects taught, and contact details.
- Timetables – Class schedules for students and teachers.
- Examination Results – Test scores, report cards, and academic progress.
- Financial Records – School fees paid, budgets, and expenses.
- Library Records – List of books, borrowers, and due dates.
- Disciplinary Records – Reports of student behavior and actions taken.

Meaning of Communication

Communication in this regard refers to the electronic transfer of data from one point to another.

Situation where communication takes place

- A teacher sends an email to students with homework instructions.
- A student attends an online class using Zoom or Google Meet.
- A school sends SMS messages to parents about upcoming events.
- Students chat in a WhatsApp group to discuss a group project.
- A head teacher uses a school management system to share exam results with parents.

Meaning of Technology

Technology refers to the scientific knowledge, skills, tools, and methods used by people to transform and use their environment

Technology refers to the use of scientific knowledge, tools, machines, or techniques to solve problems, make work easier, or improve the way we do things.

Example:

- Using a computer to type a document.
- Using a mobile phone to make a call or browse the internet.

Information and Communication Technology

Information and Communication Technology refers to the diverse set of technological tools and resources which are used to communicate, create, disseminate, store information and manage information.

ICT Tools (ICTs)

Computer (Desktop or Laptop)

A powerful digital device used to input, process, store, and display information. It runs various software applications for tasks such as word processing, spreadsheet management, programming, internet browsing, and graphic design.

Smartphone

A portable device that combines a mobile phone with advanced computing functions. It allows users to make calls, send texts, access the internet, take photos, use apps, and perform various communication and productivity tasks.

Tablet

A lightweight, touchscreen-based device larger than a smartphone but smaller than a laptop. It is used for reading digital books, accessing educational content, taking notes, browsing, and media consumption.

Printer

An output device that produces hard copies of digital documents, images, or graphics on paper. Commonly used in schools, offices, and homes for printing assignments, reports, or photos.

Scanner

An input device that captures images or text from physical documents and converts them into digital format for editing, emailing, or storing on a computer.

Projector

A display device that projects images, videos, or presentations from a computer or media player onto a larger screen or wall, often used in classrooms and meetings for group viewing.

Digital Camera / Webcam

A device used to capture still images or video. Digital cameras are used for photography, while webcams are mostly used for live video communication in online meetings or virtual classrooms.

Photocopier

A machine that quickly makes duplicate copies of documents or images, widely used in offices and schools to reproduce handouts or reports.

Flash Drive (USB Drive)

A small, portable device used for storing and transferring digital files between computers and other devices. It is convenient for backing up data or carrying work between home and school.

External Hard Drive

A large-capacity storage device used to back up or transfer large amounts of data such as videos, software, and databases. It connects to a computer via a USB or wireless connection.

Modem

A device that connects a home, office, or school to the internet by converting signals from your Internet Service Provider (ISP) into digital form that devices can use.

Router

A device that distributes internet connection from the modem to multiple devices—either through Wi-Fi or Ethernet cables—allowing phones, computers, and other gadgets to access the internet at the same time.

Interactive Whiteboard (Smartboard)

A touch-sensitive board that connects to a computer and projector, allowing users to write, draw, and interact with digital content using a pen or finger. It enhances teaching and learning through interactive lessons.

Headphones / Earphones

Audio output devices worn over or in the ears to listen to sound privately from a computer, phone, or other digital device. Useful for online learning, editing audio, or quiet environments.

Microphone

An input device that captures voice or sound and converts it into digital signals. It is used for voice recording, video conferencing, dictation, and virtual communication.

Speakers

Output devices that convert digital audio signals into sound. Used for playing music, videos, or audio instructions during lessons or presentations.

Photographic Drone

A remote-controlled flying device equipped with a camera, used to capture aerial images and videos. Common in media, surveying, security, and agriculture.

Video Conferencing Tools (e.g., Zoom, Microsoft Teams)

Software platforms that enable real-time video and audio meetings over the internet, commonly used for online classes, virtual meetings, webinars, and remote collaboration.

Email Applications (e.g., Gmail, Outlook)

Digital tools that allow users to send, receive, and organize messages and files electronically, often used for communication in academic and professional environments.

Web Browsers (e.g., Google Chrome, Mozilla Firefox)

Software that allows users to access and navigate the World Wide Web, view websites, search for information, and use online services.

Learning Management Systems (e.g., Google Classroom, Moodle)

Platforms used by schools and organizations to manage digital learning. They support uploading of materials, giving assignments, grading, and communication between teachers and learners.

UPS (Uninterruptible Power Supply)

A backup power device that provides temporary electricity to a computer or other ICT equipment when there is a power cut. It prevents data loss and gives users time to save their work and shut down safely.

Antivirus Software

A program installed on computers and mobile devices to detect, block, and remove harmful software (viruses, malware, spyware). It protects systems from being damaged or hacked.

Firewall

A security tool (either software or hardware) that monitors and controls network traffic. It helps prevent unauthorized access to or from a private network.

Surge Protector (Power Extension with Surge Control)

A device used to protect ICT equipment from damage caused by electrical surges or spikes.

Cloud Storage Tools (e.g., Google Drive, Dropbox)

Online platforms that allow users to store files safely on the internet, access them from any device, and share them with others.

Network Switch / Hub

Hardware used in offices and schools to connect multiple computers or devices within a local network so they can share files and resources like printers.

Software Applications (e.g., Microsoft Office, Adobe Reader)

Programs that help users perform specific tasks such as typing (Word), creating spreadsheets (Excel), designing posters (Publisher), or reading PDF files (Adobe Reader).

Backup Software

Used to create copies of important files and system settings, which can be restored in case of data loss or system failure.

Barcode Scanner

A device used to read barcode labels, commonly found in libraries, shops, and warehouses for tracking items and speeding up transactions.

Point of Sale (POS) Software

A system used in shops and businesses to manage sales transactions. It records what is sold, calculates the total price, prints receipts, and can track stock levels and generate sales reports.

Accounting Software (e.g., QuickBooks, Tally)

Used to manage financial records such as income, expenses, payroll, and invoices. Helps with budgeting, tax calculations, and financial reporting.

Word Processing Software (e.g., Microsoft Word, Google Docs)

Allows users to type, edit, format, and print text documents such as letters, reports, and assignments.

Spreadsheet Software (e.g., Microsoft Excel, Google Sheets)

Used to organize and analyze data using rows, columns, and formulas. Commonly used for budgeting, record-keeping, and calculations.

Presentation Software (e.g., Microsoft PowerPoint, Google Slides)

Helps users create slideshows with text, images, animations, and videos—useful for teaching, business presentations, and student projects.

Database Software (e.g., Microsoft Access, MySQL)

Stores large amounts of organized data for easy retrieval and updating. Used in schools, hospitals, and banks to manage records.

Graphics and Design Software (e.g., Adobe Photoshop, Canva)

Used for creating and editing images, posters, logos, and other visual content. Useful in marketing, education, and creative industries.

Web Design Software (e.g., WordPress, Dreamweaver, MS Expression Web)

Helps users create and manage websites without needing deep programming skills.

Learning Platforms (e.g., Moodle, Google Classroom)

Allow teachers to deliver lessons, post assignments, grade student work, and communicate with learners online.

Internet (Web) Browsers (e.g., Chrome, Firefox, Safari)

Software used to search for and view websites, download files, and access online tools.

Media Players (e.g., VLC, Windows Media Player)

Play audio and video files, useful in both entertainment and education for listening to music, podcasts, or viewing learning videos.

PDF Readers (e.g., Adobe Acrobat Reader)

Software used to view, print, and sometimes edit PDF documents—commonly used for forms, e-books, and reports.

Communication Software (e.g., WhatsApp, Zoom, Microsoft Teams)

Used for sending messages, making video/audio calls, and conducting virtual meetings or lessons.

Computer System

A computer system is a collection of computer components working together to input data, process data and output results called information.

Computer components may include hardware, software, people (users), procedures and information.

Components of a computer system

- Hardware
- Software
- Computer users/Human ware/Liveware
- Procedures
- Information

Computer Hardware

Hardware refers to the physical components of a computer system. Any component you can touch, feel and pick up is called a computer hardware

Examples of hardware computer components include; Keyboard, mouse, monitor, printer, hard drive etc.

Computer hardware includes;-

- Input devices like the keyboard and mouse,
- Processing devices like the Microprocessor Chip,
- Storage devices like the Hard disks and the CDs,

- Output devices like the monitor and the printer
- Communication devices like Modem, Router, Switch, NIC, VGA Cable, HDMI Cable etc.

Computer software

Software refers to the set of electronic instructions that tell the computer hardware what to do and how to do it.

Software is a term for electronic instructions that tell the computer how to perform a task.

These are a series of programs (instructions) that tell the computer what and how to work.

Computer software can be grouped into System software and Application software

Information

Information refers to processed data that has meaning and value to the end user. When data is processed, it gives meaning. For example, a receipt gives information about purchased items and a report card gives information about a student's performance.

Procedure

A procedure refers to a sequence of actions conducted in a certain order to accomplish a given task or make sure that a computer system run smoothly.

For example, storing data on a flash disk or external hard disk requires a series of procedures like inserting the disk into the computer, finding the information to be stored, copying it and putting it on to the disk and finally removing the disk from the computer

Users/Human ware/Liveware

Computer Users (Human ware) refers to the people who operate and initialize instructions to the computer system.

They design and develop computer systems, operate the computer hardware, create the software, and establish procedures for carrying out tasks.

Types of computer users

Ordinary user -is someone without much technical knowledge of computers but uses computers to produce information for professional or personal tasks, enhance learning, or have fun. Ordinary users include Computer students, Typists (Secretaries), etc.

Professional user -is a person in a profession involving computers who has had formal education in the technical aspects of computers; Examples include Computer programmers, webmasters, etc.

Activity: Classifying Components of the computer system

1. Classify the following computer components under the categories: hardware, software, information, procedures and people

Computer technician, flash disk, Windows 10, S.I ICT Exam, Secretary, DVD Drive, Microsoft Word, ICT user manual, creating and printing a letter, photocopier

2. Imagine you have been invited by your local council chairperson to educate people in their community about hardware and software. How would you explain the difference between hardware and software?

A Computer and its Parts

A computer is an electronic device which inputs, processes and stores data or information and outputs information when given a set of instructions. Computer instructions are known as commands.

A Computer is an electronic device that accepts data input, processes it according to some specified instructions, outputs the information and stores the results for future use.

A Computer is an electronic device which stores and processes data in binary form according to the instructions given to it in available program.

The operating speed of a computer is in Millions of Instructions per second (**MIPS**).

What a computer does?

There are four functions that a computer is built to perform

- **Input:** Computers receive data and instructions through input devices like keyboards, mice, touch screens, etc.
- **Processing:** The computer's central processing unit (CPU) interprets and executes instructions, performing calculations and manipulating data according to the program's logic.
- **Storage:** Computers store data and programs in various types of memory, such as RAM (random-access memory) for temporary storage and hard drives or solid-state drives for long-term storage.
- **Output:** After processing the data, the computer produces output through output devices like monitors, printers, speakers, etc.

Characteristics of modern computers

Speed: Computers are quite fast in their operation in that their speed is measured in Millions of Instructions Per Second (MIPS).

Diligence : This refers to the computers' ability to perform the same task over and over (repeatedly) for a long time without getting bored or tired.

Accuracy: This is the computer's ability to process large amounts of data without making mistakes. They can process large amounts of data and generate error-free results, provided the data entered is correct. It follows therefore, that if wrong data is fed into the computer, the resulting output will be incorrect. Hence the saying; Garbage In Garbage Out (GIGO).

Automation: This is the ability of computers to carry out tasks without instruction by the user. Such tasks include; automatic detection and cleaning of viruses. This is at times because of the instructions (programs) installed in them.

Storage: This is the ability of computers to keep/ retain data or information either temporarily or permanently.

Artificial intelligence: This is the computers' ability to mimic human behavior.

Versatility : This means that a computer has the ability to perform more than one task at the same time.

Communication. They can communicate with other computers to share data, information, programs and instructions through networks. Networked computers have the ability to communicate and share resources.

Basic Parts of A computer

The basic parts of a computer include the following.

The system unit

It's rectangular box that houses the internal electronic delicate parts of a computer.

The most important part is the CPU or processor which act as the brain of a computer.

Mouse

A mouse is a small device used to point and select items on the computer screen. It used in the movement of the cursor

Keyboard

A keyboard is an input device mainly used for typing text and issuing commands into the computer.

A **monitor** is an output device that displays information in visual form like text and graphics to the user.

A **speaker** is an output device that is used to play sound. They may be built into the system unit or connected with cables.

Microphone - An input device that captures sound, allowing users to record voice, speak in meetings, or give voice commands.

The Information Processing Cycle

The information processing cycle is a series of steps the computer follows to receive data and process it into information.

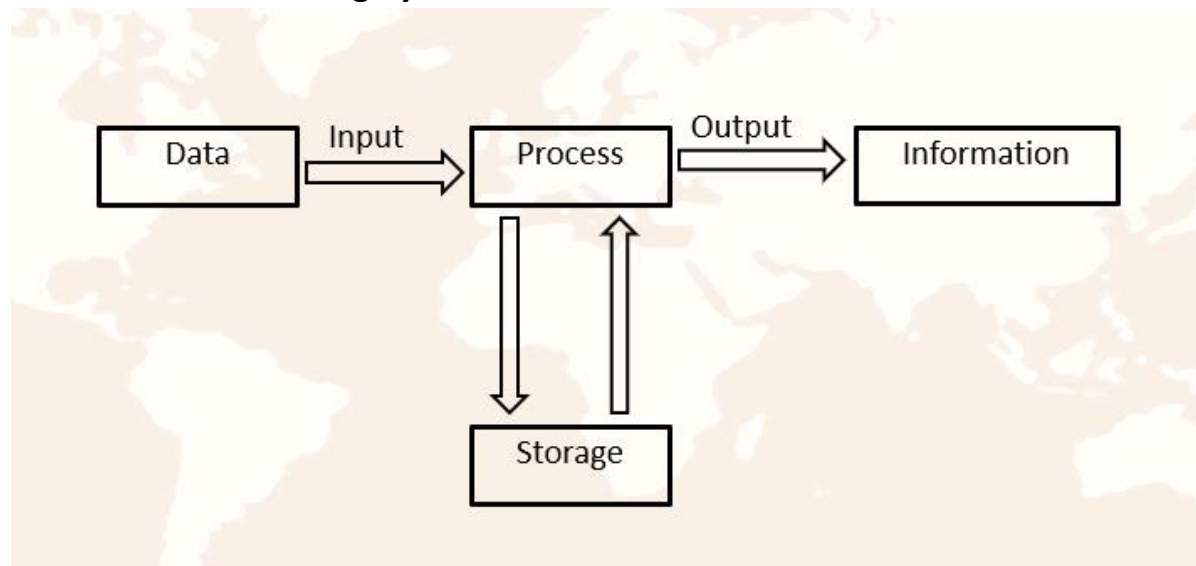
The 4 basic operations of the information processing cycle are input, processing, output and storage.

Data processing

Data processing is the activity of manipulation of raw facts to get meaningful information.

These are the steps that are taken to convert raw facts, which are data, into information.

Information Processing Cycle



Stages of Information processing cycle

Input stage

Data, instructions and commands are entered into the computer with the help of input hardware devices for processing.

Processing stage

Data is converted, changed, transformed, interpreted, manipulated into information that can be understood by the user.

Output stage.

At this stage, information is produced, conveyed or displayed to the user. The user can be able to see the results of processing with the help of output devices.

Storage stage.

Information is stored permanently for future use in the computer.

USES OF ICTs IN SOCIETY

Information and Communication Technology refers to the set of technological tools and resources used to create, communicate, store, disseminate and manage information.

Area where ict TOOLS are used

- Transport
- Education,
- Business
- Health
- security
- Politics and Governance
- Communication
- Entertainment / leisure
- Technical and scientific uses

How ICTs are applied in Transportation

- GPS and Navigation Systems: ICT tools like GPS devices and mobile apps (e.g., Google Maps) help drivers and passengers find the best routes, avoid traffic, and reach destinations faster.
- Traffic Management Systems: Traffic lights, road sensors, and surveillance cameras use ICT to monitor and control traffic flow in busy areas, helping to reduce congestion and accidents.
- Electronic Ticketing and Payments: Buses, trains, and airlines use ICT systems for booking tickets online, checking in, and making cashless payments using cards or mobile money.
- Fleet Management: Transport companies use ICT tools to track their vehicles, monitor fuel use, plan deliveries, and communicate with drivers in real time.
- Public Transport Information Systems: Passengers can access information about bus or train schedules, arrival times, and delays through websites, mobile apps, or digital signboards.
- Security and Surveillance: CCTV cameras and tracking systems are installed in stations, airports, and vehicles to monitor activities and ensure safety.
- Vehicle Safety Features: Modern vehicles use ICT-based systems like automatic braking, lane assist, and collision alerts to reduce the risk of accidents.

Application of ICTs In the Area of Education

- Computers can be used to help students learn a new subject on their own through Computer Aided Instructions (CAI)
- Students use computer software to learn at their own pace (speed) and answer questions through Computer Assisted Learning (CAL)
- Teachers can use computers to mark and assess the students' performance through Computer Assisted Assessment (CAA). This reduces the time and labor spent marking the students' scripts.
- Schools use computers to create school websites for sharing information with the public.
- Productivity tools like desktop publishing and presentation software are used in projects and other school activities.
- Computers are used for calculating mathematical arithmetic by students and teachers in class.
- With Use of School Administration and Management Systems. (SAMS) Records management is made easier because all details of learners can be held on computer, and easily retrieved, reducing administrative costs.
- Students' Progressive Report Cards can be produced electronically by use of computers instead of hand written ones.

- Distance learning through computer-based training. People get award such as degrees without going to class.
- Teachers use simulation software to perform difficult or dangerous experiments in class.
- Use of special facilities for students with disabilities like text to speech and speech recognition to help blind students.

Application of ICTs In The Area Of Business

- Computers enable people to Work from home, using a computer connected to the employer's network or via the Internet. This is known as **Telecommuting**.
- Computers have created more jobs such as Computer technicians, Computer teachers, etc.
- Buying and selling Computers and its components is a source of income to individuals, and companies.
- Through, Computer Aided Design (CAD), scale drawings, and excellent designs can be created easily.
- Computers are used for sending and receiving Mobile Money and making worldwide money transfers.
- Banks use Computers to manage transactions and Automated Teller Machines ATMs for 24 hours banking.
- Computers help in Business Advertisement through creating websites, internet, flyers, brochures and billboards.
- Computers are used in typesetting business for production of document printouts and publication of Books for sale.
- Computers are used for E-Commerce: the sale of goods and services over the internet.

Applications of ICTs in the Area of Health

- Hospitals use computers for managing and storing Records electronically, rather than paper files.
- Hospital Administration is also aided by printing labels, allocating beds, make appointments, staff rotas, etc.
- Internet helps us get Web sites for information on health care, treatments, conditions, etc.
- Monitoring/Diagnosis such as Heart rate, blood pressure, etc. is aided by Computer Expert systems.
- Medical Training is facilitated by Simulation software and on-line data sources.
- Computers are used to carry out many surgical procedures such as laparoscopic surgeries
- They are used in diagnosis and cure of many disease for example CT Scan, Ultra sound devices and Magnetic imaging.(MRI)
- They enable online consultations by medical professionals.
- Use of computer assisted tests can be carried out before prescribing treatment.
- Use of computer assisted life saver machine.
- Enable faster communication between patients and doctors
- Can be used to monitor the patients in hospitals e.g. CCTVs

Application of ICTs in the Area of Security

- Computers aid monitoring security through cameras, Automatic number plate recognition, etc.

- Communication systems are widely used in the military to coordinate the personnel.
- Some computer systems can detect temperatures and alarm in case of danger of fire outbreaks.
- Computers are used for capturing data for Police National Computer Databases —, vehicle number plates, criminal's fingerprints, etc.
- Computers are used to detect presence of illegal devices such as bombs.
- Computers are also used for controlling dangerous weapons such as missiles.
- Computers are used for storing criminal databases and information at Police stations.

Uses of ICTs in the Area of Politics and Governance

- Paying government tax online through a government website.
- Online forms such as vehicle registration and passport forms.
- Advertising government tenders and Applying for government tenders
- Public records -A maintained database of statistical information such as electoral register and census data can be availed online.
- Use of electronic voting during elections.
- Government departments can use a computer-based platform to get feedback from the citizens.

Uses of ICT in Art, Leisure and Entertainment

- Computers enables people to play computer games like GTA Vice City, Need for Speed, Solitaire etc.
- Internet has promoted social networking that has enabled interaction between people. Examples include Twitter, Facebook, LinkedIn etc.
- Computers can be used to play music during free time.
- Computers can be used to watch movies and videos.
- Digital cameras can be used to record and capture videos on parties.
- Computers are used to compose and edit songs by producers through using audio and video production software.
- Internet enables people to read magazines online.

Uses of ICTs in the Area of Communication

- E-mail: Electronic Mail sent from one person to another using connected computers helps a lot in the area of communication.
- Video Conferencing enables people in different locations to conduct meeting as if they are in the same location.
- Computers are used for Faxing: Sending an image of a document electronically.
- Computers enable people to send voice, image, text and data though telephones and mobile cell phones.
- Social Networks such as Facebook, and Twitter enable people to stay in touch with their relatives, friends and interests.

Technical and Scientific Uses of ICTs

- In Astronomy, Computers are essential tools to study the behavior of the complex systems in space as regards to their movements, interactions etc.
- Through Computer Aided Manufacture (CAM), computers can be used to control the production of goods in factories.
- Monitoring highway traffic

- Computers are used to tell schedules of water vessels, train, buses to their respective stations.
- Computers are used very extensively in design of roads. Roadways and bridges are designed using software programs like CAD etc.

Application of ICTs in Security

- One major application is in surveillance systems, where Closed-Circuit Television (CCTV) cameras are used to monitor and record activities in real-time, helping to prevent crime and gather evidence.
- ICT is also used in access control systems, such as biometric scanners that use fingerprints, facial recognition, or ID cards to restrict access to sensitive areas like offices, banks, or data centers.
- Alarm systems powered by ICT detect unauthorized entry or unusual movement and alert security personnel.
- Cybersecurity tools like firewalls, antivirus software, and encryption help protect computer systems and digital information from hackers, viruses, and data breaches.
- ICT also enables quick and effective communication among security teams through mobile phones, radios, and walkie-talkies, especially during emergencies.
- GPS tracking systems are used to monitor the movement of vehicles or people, which is particularly useful in law enforcement or logistics.
- ICT supports data storage and retrieval, allowing authorities to maintain databases of criminal records, fingerprints, or surveillance footage for investigation and legal purposes.

COMPUTER HARDWARE

Computer hardware refers to the physical and tangible components of a computer system which you can touch, feel and pick up.

Categories of Computer hardware

- Input devices
- Processing devices
- Storage devices
- Output devices
- Communication devices

Input devices

Input devices are hardware components used to enter/feed data and control signals (instructions) into a computer system for processing.

Examples of Input Devices

Input Device	Function
Keyboard	Enters text, numbers, and commands.
Mouse	Controls the pointer and selects items on the screen.
Scanner	Converts physical documents/images into digital format.
Webcam	Captures live images or video.
Microphone	Inputs data in form of sound
Joystick	Controls movement in games and simulations.

The Keyboard

A keyboard is an input device, consisting of a set of keys (buttons) used to input text and commands into a computer.

Parts of a Keyboard

Functional keys: These are keys found across the top row of the keyboard. Each key is composed of letter F and a number ranging from 1 to 12. Each functional key performs a specific operation based upon the software being used.

Numeric keypad: This is an extra section to the right of the keyboard.

Navigation/Cursor/Arrow keys: These keys are used to navigate through documents and websites. They include the up, down, left, and the right arrows.

Typewriter Keypad (Alphanumeric Keypad): This consists of the keys in the center of the keyboard. It has buttons with alphabetical letters and numbers hence the name —alphanumeric keypad. Most of the typing is done with these keys.

Special Keys: These are the keys that are used to perform specific functions in the computer. They include; Shift key, Tab key, Control key, Caps Lock, Num Lock, Backspace key, Enter key, Delete key, Alt key, etc.

Functions of some selected keys on the keyboard

Backspace Key - Deletes the characters to the left of the cursor (or insertion point) and moves the cursor to that position. Or Ctrl+ Backspace key removes the whole word to the left of the cursor.

Caps Lock Key - A toggle key that, when activated, causes all alphabetic characters to be uppercase. - To facilitate continuous typing in upper case when activated.

Delete Key - To remove the character, space, text or word to the right of the cursor position. Ctrl+Del removes the whole word to the right of the cursor position.

Shift Key

- Activates second function of different keys.
- Activates sticky keys
- Selects files and folders in specific order
- Modifies function Keys e.g. Shift + F1 = F13 and Shift + F2=F14.
- Anchors the insertion pointer in word processors

Ctrl key – Short for Control, this key is used in conjunction with other keys to produce control characters. The meaning of each control character depends on which program is running.

Enter Key – Used to enter commands or to move the cursor to the beginning of the next line. Sometimes labeled Return instead of Enter.

Esc Key – Short for Escape, this key is used to send special codes to devices and to exit (or escape) from programs and tasks.

Alt key – Short for Alternate, this key is like a second control key.

Space bar

- To create space .
- Alignment
- Moving the cursor to the next line.
- Pausing music in some music programs e.g. Power DVD,
- Playing games e.g. in pinball
- Resizing or closing the current window with a modifier key e.g. Ctrl Key +Space bar
- Helps in scrolling in some web browsers when used with Shift Key.

Modifier keys

- A modifier key is a key that modifies the action of another key when the two are pressed together. Modifier keys are commonly used to enter keyboard shortcuts.
- Common modifier keys include Shift, Control, Alt, Command, Option, Windows and Function (Fn).

Advantages of using keyboards

- They are common, commonly supplied with computers
- Entering data and commands with a keyboard is faster
- Keyboards are more reliable
- They are relatively cheap

Demerits of keyboard

- It takes a lot of time to practice in order to type quickly
- Keys can easily become faulty due to dust.
- Constant use of keyboards may cause finger injuries

The Mouse

The mouse is a hand-held pointing device. A pointing device is a special data input device, which allows users to move a pointer and make selections on the computer screen.

A ball under the mouse senses movement while the buttons on top are used for left, right and double clicking and scrolling.

An optical mouse uses a light-emitting diode and photodiodes to detect movement relative to the underlying surface.

Functions of a mouse

- Clicking: pressing and releasing the left-hand button of the mouse once.
- Right clicking: pressing and releasing the right mouse button once
- Double clicking: pressing the left button in quick succession
- Dragging: pointing at an item and then hold down the left button as you move the mouse.
- Scrolling: navigating the window page up or down by moving the wheel like button.

Other pointing devices include:

- **Stylus pen & digitizing tablet:** Stylus pen- The pen lets you draw on what is called a digitizing tablet that mirrors the surface area of the computer screen. The pen can be used as a standard mouse (without wires connected to it) or also as a free-flowing drawing device. The pen is useful for drawing since drawing graphics with a mouse tends to be inaccurate.
- **Cordless Mouse :** A Cordless Mouse connects to the computer using wireless technology, giving flexibility to the user and to move the mouse over a wider working area. This mouse runs on a battery. When you move the mouse it sends an infrared beam to a sensor which interprets it causing the pointer to move.
- **Trackball:** The trackball uses the same principle as the mouse except that the rollers are reversed, the ball is on top. It can remain stationary on your desk.
- **Touchpad:** The touchpad has sensors that sense a user's touch and send a signal to the computer.
- **Touch Screen:** A touch screen is a touch-sensitive input and display device. Users can interact with these devices by touching areas of the screen.

- **A Track Point:** A Track Point, also called a pointing stick, is a pointing device located in the middle of the keyboard between the G, H, and B keys. The Track Point is operated by pushing in the general direction the user wants the cursor to move. Increasing pressure causes faster movement.
- **Joystick:** A Joystick consists of a stick that pivots on a base and reports its angle or direction to the device it is controlling. Joysticks are often used to control video games, and usually have one or more push-buttons whose state can also be read by the computer.

Image scanner: A scanner converts hardcopy documents, drawings, or pictures to an electronic version (softcopy), which can then be stored on a disk.

Digital Camera: A digital camera allows users to take pictures and store the photographed images digitally instead of storing on a traditional film. Users transfer a copy of the stored pictures to the computer or printer by connecting a cable between the digital camera and the computer.

Digital video (DV) camera: *A digital video (DV) camera, by contrast, records video as digital signals instead of analog signals.*

After saving the video on a storage medium, users can play it or edit it and burn it to a DVD using software programs on the computer.

Camcorder: This is a light weight video camera that records data in digital form onto a storage device such as a videotape.

Webcam: *A Web cam, also called a PC video camera, is a type of digital video camera that usually sits on top of the monitor. Some laptop computers have built-in Web cams.*

Webcams enable users to capture video and still images and make video telephone calls

Microphones: A microphone is an instrument for converting sound waves into electrical energy variations, which may then input into the computer for processing, recording or audio playback. Microphones are connected to the sound card in the system unit.

Barcode Reader: *A bar code reader, is an optical reader that uses laser beams to read bar codes that are printed on items usually in super markets.*

A bar code normally consists of a unique set of vertical lines and spaces of different widths which represent coded information about the item it represents.

Magnetic Strip Card Reader: A magnetic stripe card reader reads the magnetic stripe on the back of credit cards, bank ATM cards, and other similar cards.

Exposure to a magnet or magnetic field can erase the information and contents of a card's magnetic stripe.

Remote controllers - devices that emit a beam of infrared light, which carries data signals, commonly used with TVs and modern laptops.

Sensors - Various sensors can be used to measure heat, light, pressure, acidity, oxygen concentration, water flow, metals, alcohol, etc.

Biometric Devices

A biometric device is the one used to identify a person by the measurement of his/her biological features. Every person in the world has unique biological features and some of these include fingerprints, facial images, iris (retina) prints and voice recognition.

Examples of Biometric Devices

- **Face scanner:** Biometric face scanners identify a person by taking measurements of a person's face.
- **Retina or Iris scanner:** These are devices that use *iris recognition technology* to read patterns in the iris of the eye.
- **Voice scanner:** A voice is a device that mathematically breaks down a person's voice to identify him/her.
- **Fingerprint scanner:** A fingerprint scanner captures curves and indentations of a fingerprint.

Terminals

These are computer devices that allow users to send data to or receive information from a host computer. These include:

Electronic Point Of Sale (EPOS) – a machine from where customers can pay for goods or services from. It records purchases, processes credit or debit cards and updates inventory. They are commonly used in super markets and large shopping halls.

Advantages for EPOS

- **Speed of Service:** EPOS systems streamline the checkout process, reducing transaction times and allowing for faster customer service.
- **Inventory Management:** Automated inventory tracking helps in maintaining optimal stock levels, reducing the chances of overstocking or stock outs.
- **By automating the checkout process,** EPOS systems minimize human errors in transactions, such as incorrect pricing or calculation mistakes
- **Real-Time Data:** Accurate, real-time sales data helps in making informed business decisions quickly.
- **Faster checkout and various payment options** improve the overall shopping experience.
- **Reduced Labor Costs:** Automation reduces the need for manual labor, allowing staff to focus on more critical tasks.
- **Efficient Resource Use:** Better inventory management reduces waste and improves cash flow
- **Customer Insights:** Data collected from EPOS systems can be used to understand customer preferences and tailor marketing strategies accordingly.
- **Promotions and Discounts:** Easy management of sales promotions and discounts encourages customer spending and loyalty.

Automatic Teller Machines (ATM) – a self-service machine from where bank customers can withdraw cash, deposit money, transfer funds, or inquire about their account balances. The user Inserts a bank card into the machine. The machine prints a receipt when transactions are over.

Categories of Input devices

- | | |
|--------------------------|-----------------------------------|
| • Text input devices | • Audio input Devices |
| • Pointing input Devices | • Biometric input Devices and |
| • Imaging input Devices | • Other Specialized input devices |
| • Gaming input Devices | |

Activity: Write about and list examples for each of the above different categories of input devices.

Output Devices

Output devices are hardware components that convey/display information from a computer to the user or another device. An output device is any type of hardware component capable conveying information to one or more people.

Categories of Computer output

- Text
- Graphics
- Audio
- Video

Examples of Output Devices

Output Device	Function
Monitor	Displays visual output from the computer.
Printer	Produces a hard copy of digital documents/images.
Speakers	Play sound from the computer.
Headphones	Provide personal audio output.
Projector	Projects computer display onto a larger screen.

Common Output Devices

Monitors

There are two main types of display monitors: Cathode ray tube (CRT) and liquid crystal display (LCD) monitors.

The CRT monitor works like a television because it also contains a cathode ray tube. The front of the CRT is the screen, which is coated with tiny dots of phosphor material.

The LCD monitor uses liquid crystals to create images on the screen. LCD monitors produce color using either passive matrix or active-matrix technology.

Advantages of LCD monitors over CRT

- LCDs are much thinner and weigh significantly less than CRTs, making them easier to move and requiring less desk space.
- LCDs use less electricity than CRTs, making them more energy-efficient and cost-effective in the long run.
- LCDs provide sharper images and better support for higher resolutions, which is ideal for reading text, watching videos, or editing graphics.
- Unlike CRTs, LCDs do not flicker, which reduces eye strain and makes them more comfortable for long-term use.
- LCDs contain fewer hazardous materials (e.g., no leaded glass), making them more eco-friendly and easier to recycle.
- LCDs emit less heat compared to CRTs, which contributes to a cooler working environment.
- LCD technology supports much larger screen sizes without increasing bulk, making it ideal for widescreens and HDTVs.

Projectors

A data projector takes the image that displays on a computer screen and displays it onto a large screen or wall so that an audience of people can see the image clearly. Projectors can be specified according to various factors including: picture brightness, lamp life, ability of the lens to be adjusted, ability to accept different computer resolutions and the zoom ratio.

Printers

A printer is an output device that produces text and graphics on a physical medium such as a paper. A print out is often referred to as hard copy.

Factors to be considered when buying a printer

- **Printing speed:** Defines the number of pages to be printed in a minute.
- **Internal memory or buffer:** This defines how much print data can be queued by the printer.
- **Networking:** This defines whether it can be used as a network printer
- **Duplexing:** This defines whether the printer can print on both sides of the paper.
- **Operating system compatibility**
- **Print resolution (dots per inch):** Defines the quality of text printed on paper.
- **Print technology:** Defines whether the printer uses toner or cartridge ink

Audio output devices

Audio output devices are the components of a computer that produce sound. Two commonly used audio output devices are speakers and headsets

Interactive board or smartboard

It is a large interactive whiteboard which enhances learning and teaching by using touch technology.

Plotter

It is a special printing device used to produce high quality line drawings such as building plans, electric circuits, maps and charts

Storage Devices

Storage devices are used to save/store/keep data and programs permanently or temporarily for retrieval and use.

Examples of Storage Devices

Storage Device	Function
Hard Disk Drive (HDD)	Stores operating systems, software, and files permanently.
Solid State Drive (SSD)	Faster storage with no moving parts, used for operating systems and apps.
USB Flash Drive	Portable device for transferring and storing files.
Memory Card	Stores data in mobile devices and cameras.
CD/DVD	Stores data, music, or videos for reading or writing via a disc drive.

Classification of Storage Devices

Computer storage devices may be classified as primary storage or and secondary storage.

Primary memory refers to internal memory directly accessible to the CPU. It consists of both Volatile memory such as RAM and Non-Volatile memory such as ROM.

Secondary storage is used by Computer systems to store larger amounts of data, and information more permanently than allowed with primary memory.

Primary storage

Primary storage, also known as main memory, is the part of the computer where data and programs currently being used are stored temporarily. It is directly accessible by the Central Processing Unit (CPU) and is essential for the computer to function.

Key Characteristics of Primary Storage

- **Volatile:** Loses all data when the computer is turned off.
- **Fast access:** Much faster than secondary storage.

- Limited capacity: Usually smaller in size compared to secondary storage.
- Temporary: Stores data only while it is needed for processing.

Types of Primary Storage

Random Access Memory (RAM)

RAM is a type of primary storage that temporarily stores data and instructions the CPU needs while the computer is running.

Read Only Memory (ROM)

ROM is a type of primary storage that permanently stores important instructions that do not change, such as those needed to start the computer.

Differences between RAM and ROM

RAM	ROM
Volatile, temporally	Non -Volatile, permanent
Contents lost when power goes off	Contents remain when power goes off
Read and Write	Read Only
Can be increased	Can't be Increased
Data Not recorded at Factory	Firmware Recorded at Factory

Differences between Primary storage and Secondary storage

Primary storage	Secondary storage
Primary storage is closely connected to the CPU and is mainly used for temporary storage of data and instructions during processing	Secondary storage is relatively permanent but further away from the CPU
Primary storage is faster to access than secondary storage	Secondary storage is relatively slower to access than primary storage
Primary storage is mainly volatile (i.e., RAM contents are lost when power is turned off)	Secondary storage is non-volatile
Limited storage capacity	Large storage capacity
Holds data and instructions currently in use	Stores data and programs not in immediate use

The purpose of Primary storage or RAM in a computer

- RAM temporarily stores data while it is being processed (working area)
- Part of RAM is used by Operating system to control computer operations
- Part of RAM is used to store all the application programs being used by the users
- RAM is an output area where results of processing are output before being stored permanently.

Secondary Storage

Secondary storage refers to non-volatile storage devices used to store data and programs permanently. Unlike primary storage (RAM), secondary storage retains data even when the computer is turned off. It is used for long-term storage of files, applications, and the operating system.

Key Terms used

A **storage medium** (media is the plural) is the physical material on which a computer keeps data, instructions, and information. Examples of storage media are hard disks, solid state drives, memory cards, USB flash drives, ExpressCard modules, optical discs, smart cards, magnetic stripe cards, and microfilm.

A **storage device** is the computer hardware that records and/or retrieves items to and from storage media.

Writing is the process of transferring data, instructions, and information from memory to a storage medium.

Reading is the process of transferring these items from a storage medium into memory.

Storage capacity refers to the maximum amount of data a storage device or medium can hold

Categories of Secondary Storage Media

There is a wide variety of storage devices in the following categories.

- Magnetic media
- Optical media
- Solid-state media

Magnetic media

Magnetic storage media represent data as magnetic spots on the tape or disk, with a magnetized spot representing a 1 bit and the absence of such a spot representing a 0 bit.

Common examples of magnetic media include:

- Magnetic tape
- Floppy disk,
- Hard disks
- Magnetic stripe Card

Optical media

Optical media refers to storage media that uses laser technology to read and write data. Data is stored on the surface of the disc in the form of tiny pits and lands, which a laser beam scans to retrieve or record information.

The reflected light is converted into a series of bits that the computer can process.

Examples of optical media

- CD (Compact Disc)
- DVD (Digital Versatile Disc)
- Blu-ray Disc

Care for optical disks

- Always hold the disc by the edges or the center hole.
- Avoid touching the shiny surface where data is stored.
- Do not bend or apply pressure to the disc.
- Store discs in protective cases or sleeves to prevent scratches and dust.
- Keep discs in a cool, dry place – away from direct sunlight and heat sources.
- Store discs vertically (upright) like books, not flat or stacked.
- Avoid exposure to magnetic fields, extreme humidity, or temperature changes.
- Do not write on the disc with sharp pens or markers – use only soft-tip markers.
- Avoid sticking labels on the disc as they can cause imbalance.
- Keep discs away from food and liquids.

Solid-state media

Solid-state media consist entirely of electronic components and contain no moving parts.

Solid state is a non-volatile storage that employs integrated circuits rather than mechanical, magnetic or optical technology

Examples of Solid-state media

- Solid-State Drive (SSD)
- USB Flash Drive
- Memory Cards (e.g., SD, microSD cards)

Processing devices

Processing devices are the core components that interpret and execute instructions to process/transform data into Information.

Processing of information that takes place in a computer is mainly handled by the CPU. The CPU, like many other parts, is housed in the system unit and connected to the motherboard.

The CPU works with primary memory or RAM. RAM serves as a temporary storage area for data and the program instructions needed for processing. Information for processing must be transferred into memory from the secondary storage devices for easy access by the CPU. Processor speeds are measured in cycles per second or hertz e.g., 1500Hz.

Note:

1000Hz = 1KHz	1000KHz = 1MHz	1MHz = 1Ghz
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Examples of Processing devices

Processing device	Function
CPU (Processor)	Executes instructions and manages data processing.
GPU	Handles complex graphics and video rendering tasks.
Motherboard	Connects all computer parts and allows communication between them.

Difference between CPU and system unit

A system unit is a computer case housing all the internal components of the computer whereas CPU refers to the device that does all the processing requirements of a computer.

Communication Devices

Communication devices are used to send and receive data between computers and networks. In a computer system, the different hardware devices must communicate, for data or information to move from one device to another. This communication can be internal within a computer system or external between computer systems. Internally, computer hardware components are connected through cables referred to as buses. Externally, devices are connected through cable such as VGA, HDMI etc.

Examples of Communication devices

Communication Device	Function
Network Interface Card (NIC)	Connects a computer to a local area network (LAN).
Modem	Converts digital signals for internet access over telephone lines.
Router	Directs data between networks and connects multiple devices.
Hub	Connects devices that use the same protocols on the network
Switch	Joins the computers on the same network
Wi-Fi Adapter	Enables wireless network connection.

Computer Peripheral Tools

Computer peripheral tools (or simply peripherals) are external devices that are connected to a computer to either input data, output data, or extend the computer's capabilities.

Examples of Computer Peripherals

- Scanners
- Keyboards
- Printers
- Mice
- Plotters
- Flash disk
- External hard drive
- Projector

Peripheral Interfaces and Ports

In computing, an interface is a shared boundary across which two or more separate components of a computer system exchange information. Common interfaces include Video Graphics Array (VGA), Parallel, Serial, Personal system 2(PS/2), Infrared, Bluetooth, power interface, etc.

A **port** is a physical or virtual connection point on a computer where peripheral devices can be plugged in to send or receive data.

Examples of Computer Ports

- **USB (Universal Serial Bus) Port:** Used for flash drives, keyboards, mice, printers, external hard drives, etc.
- **Audio Jack (3.5mm port):** Used for headphones, microphones, and speakers.
- **HDMI (High-Definition Multimedia Interface) Port:** Used for connecting monitors, TVs, and projectors with video and audio.
- **VGA (Video Graphics Array) Port:** Used for connecting older monitors and projectors (video only).
- **DisplayPort:** Used for high-resolution monitors (alternative to HDMI).
- **Ethernet Port (RJ-45):** Used for wired internet/network connections.
- **Parallel Port:** Used for old printers and scanners. A parallel port sends multiple bits of data simultaneously over multiple wires.
- **Serial Port (RS-232):** Used for older modems and industrial devices. A serial port sends data one bit at a time over a single wire.
- **PS/2 (Personal System/2) Port:** Used for connecting older keyboards and mice.
- **FireWire (IEEE 1394) Port:** Used for high-speed data transfer from camcorders, external drives
- **Thunderbolt Port:** Used for high-speed data transfer, displays, and power (common on Apple devices).
- **Power Port:** Used for supplying power to laptops or desktops.

Computer System Assembly and Start-up

Assembling means to put together different components of something.

Computer system assembly refers to the process of physically putting together the individual components of a computer.

While technical people can assemble all the parts inside the system unit, you will use a pre-assembled system unit and connect it with other parts of the computer.

It is important to understand the importance of each part of the computer that you will connect and the port where it is connected. Port is a technical term that means an opening on the system unit where you can connect a peripheral device to work with the computer.

Some of the parts of the computer you may need to assemble to enable the computer to function include the following:-

- Mouse
- Keyboard
- Monitor
- Power cables
- System unit and
- UPS (Uninterruptible Power Supply)

Booting

The process of starting a computer and loading an operating system into memory to make the computer ready for use.

Types of Booting

Cold Booting

Cold booting refers to the process of starting a computer from a completely powered-off state. It occurs when the system is turned on after being shut down, where the power supply is activated and the hardware components undergo initialization.

Warm Booting

Warm booting refers to the process of restarting a computer which has been running or operating. This typically occurs when the operating system is restarted using a command, such as selecting "Restart" from the operating system's start menu or pressing the **Ctrl + Alt + Delete** keys in Windows.

Reasons for warm booting

- Before and after installation of a new application or utility software
- After uninstalling an application or utility software
- When a peripheral, hardware component fails to function
- After uninstalling or disconnecting a hardware component
- During and after installation of operating system
- During installation of a new computer hardware
- After configuring a server on a network or changing network settings
- After changing system control settings
- When a user wants to clear a malicious infection like malware, spyware, viruses from the memory
- When switching from one operating system to another – multiple OS system
- When the computer stops responding to commands (freezes or deadlocks)
- After malware or virus scanning
- When there is suspected system attack or tapping
- When the computer system slows down
- When application software fails to load or work
- When power voltage suddenly lowers

Benefits of restarting a computer

- It flushes RAM and removes other files that could be causing computer freezing.
- Fast Performance. Reboots are known to keep computers running quickly.
- Stops Memory Leaks. These occur when a program doesn't close properly.
- Fixes Internet Connection.
- Recovers the computer from errors

Step-by-Step Process of Booting

- When the computer is powered on, the power supply activates and sends an electrical signal to the motherboard and other connected devices, such as the RAM, hard drive, and peripherals, preparing the system for booting.
- The CPU resets to prepare for the boot process. It then looks for the ROM (Read-Only Memory) where the BIOS (Basic Input/Output System) is stored, as it manages hardware initialization.
- The BIOS performs the Power-On Self-Test (POST), a diagnostic routine that checks essential hardware components such as RAM, CPU, keyboard, storage devices, and graphics card to ensure they are properly connected and functioning.
- The POST results are compared with data stored in the CMOS (Complementary Metal-Oxide-Semiconductor) chip, which holds the system configuration and settings. Any discrepancies or errors will result in an error message or beep codes to alert the user.
- If the hardware checks pass, the BIOS searches for the bootloader program, which is typically located on the primary storage device (like an HDD or SSD).

- The bootloader is loaded into memory and executed. It then loads the kernel of the operating system into RAM. The kernel is the core component that manages hardware resources and enables software to interact with the system.
- Once the kernel is loaded, the operating system continues to load configuration files, device drivers, and other system components into RAM.
- Once everything is initialized, the desktop and icons appear on the screen, indicating that the computer is ready for use.

Steps to assemble a working computer given system unit, flat screen monitor, keyboard, mouse, VGA cable, power extension cable, UPS, Printer, printer data cable and 3 power cables.

- Connect the mouse to the PS/2 mouse port or USB port
- Connect the keyboard to the PS/2 keyboard port or USB port
- Connect the printer to the LPT port using the available printer data cable.
- Connect the first power cable to the power port on the printer to draw power from the power extension cable.
- Connect the second power cable to the power port behind the system unit to draw power from the UPS.
- Connect the third power cable to the power port at the base of the flat monitor to draw power from the UPS.
- Connect the VGA cable from the monitor to the video port behind the system unit
- Connect the UPS to draw power from the extension cable.
- Connect the power extension to the wall socket
- You can finally switch on the power source for the extension cable

Procedures to start a computer properly

- Switch on the wall socket
- Switch on the power extension cable, power stabilizer and the UPS in the same order.
- Switch on the monitor
- Switch on the printer
- Lastly switch on the system unit

Steps to Shut down a computer properly in Windows 10

- First close all programs that are running (close active windows)
- Click the Windows button (or Start button) to get system menu
- On the system menu, click power button
- Select Shut Down Command. The computer will display the screen preparing to shut down. The computer finally, turn off the system unit
- Switch off the monitor
- Switch off the power source (I.e., UPS, power extension and wall socket)

FILE AND FOLDER MANAGEMENT

Files

A file is a collection of related data or information stored together on a computer under a specific name called file name.

A file is a basic unit of storage in a computer identified by a unique name.

Operation performed on a file

Computer users and programs can perform several basic operations on files. These operations allow for effective management, storage, and use of data. File operation include

- Creating files
- Moving(Dragging from one location to another or cut)
- Copying
- Selecting single or multiple files (ctrl + click / Shift + click)
- Renaming
- Searching and Sorting by name, date or file extension (type)
- Printing
- Deleting
- Restoring

How to create a new file

Depending on the programs installed in your computer, you can create different types of files such as drawings, text document etc.

To create a new text document:

- On the free space on the desktop, right click.
- Point to new, click Text Document from the list available on application.
- Type a new name for the new file to replace the temporary name and press Enter key.
NB: In windows, file name can contain up to 255 characters, including spaces but, with no special symbols such as \/:*?"<>.

Selecting a Single File

Using the Mouse, Click once on the file you want to select. It will be highlighted to show it is selected.

Selecting Multiple Files

A. Selecting Files That Are Next to Each Other (Contiguous Selection)

Using the Mouse:

1. Click the first file.
2. Hold down the **Shift** key.
3. Click the last file in the group.
4. All files between the first and last will be selected.

Using the Keyboard:

1. Select the first file using arrow keys.
2. Hold down **Shift** and use arrow keys to extend the selection.

B. Selecting Files That Are Not Next to Each Other (Non-Contiguous Selection)

Using the Mouse:

1. Hold down the **Ctrl** key.
2. Click each file you want to select, one by one.
3. Release **Ctrl** when done.

Using the Keyboard:

Use Ctrl + Arrow keys to move, and Ctrl + Spacebar to select files one by one.

C. Selecting All Files in a Folder

Keyboard Shortcut:

Press Ctrl + A to select all files in the current folder.

Sorting files

- Right-click any open space within Windows Explorer and select Sort By.
- Choose to sort by Name, Date modified, Type, or Size.

- To view more sorting options, click More

Copying files

Cut or copy command are used to move or create a duplicate of an item respectively. When you cut or copy an item, it is temporarily held in a temporary storage location known as the clipboard.

Folders

A Folder is a virtual container for storing files and subfolders in the computer.

A folder is a named storage location where related files can be stored.

A folder is also known as a **directory** in some operating systems and has the following features / properties:

- A folder has a name
- A folder has a path originating from a special directory called root directory
- A folder has special access permissions for authenticated users.
- A folder or directory may be created inside another folder or directory. Such a folder or directory is called a subfolder or a subdirectory.

Creating a new folder

To create a new folder:

- Using my computer icon, on the folder tree on the left pane, select the location (desktop) in which you want to create a new folder.
- On the Home menu tab, Click new folder
- Type a new name for the folder to Replace the temporary name, then press Enter key.

Alternatively

- Right Click on the free space on the desktop,
- Point to New, click folder. Type a new name for the folder and press Enter

To copy a file or folder using File Explorer:

- Using my computer icon, display the Explorer window.
- Select the file or folder to be copied.
- On the Home menu, click copy.
- Select the drive or folder where you want the item to be copied.
- From the Home menu, click paste.
- Information or item is pasted to a new location.

Alternative Method (Using Right-Click):

- Right-click the file or folder → choose Copy.
- Go to the destination → Right-click an empty area → choose Paste.

Deleting Files and Folders

In Windows, when you delete an item from the hard disk, it is temporarily held in a special folder called the Recycle bin from where it can be restored if necessary.

Warning: Items deleted from a removable storage are not held in the recycle bin and are completely lost.

Steps to delete a file or folder

- Using This PC or File Explorer icon, display the Explorer window.
- Select the file or folder to be deleted.
- Right-click on the selected item.
- From the pop-up menu, click Delete.
- If prompted, confirm the deletion by clicking Yes
- The item is moved to the Recycle Bin.

Renaming files and Folders

Renaming refers to changing the previous name to a new name.

Steps taken to rename a file or folder

- Using **This PC** or **File Explorer** icon, display the Explorer window.
- Select the file or folder to be renamed.
- Right-click on the selected item.
From the pop-up menu, click **Rename**.
- Type the new name for the file or folder.
- Press **Enter** to apply the new name.

Restoring deleted files and folders

- Double-click the **Recycle Bin** icon on the desktop to open it.
- Browse and select the file or folder you want to restore.
- Right-click on the selected item.
From the pop-up menu, click **Restore**.
- The file or folder is restored to its original location.

Emptying the recycle bin

- Right-click on the Recycle Bin icon on your desktop.
- Select Empty Recycle Bin from the context menu.
- A prompt will appear asking if you are sure you want to permanently delete the items.
Click Yes to confirm.

Moving a folder

To move a folder to a desired location:

- Click the folder and hold
- Drag and drop to the desired location

Alternatively

- Select the file or folder you want to move.
- Right-click on the selected item.
- From the pop-up menu, click Cut.
- Navigate to the destination folder or location where you want to move the file or folder.
- Right-click in the destination folder or location.
- From the pop-up menu, click Paste.
- The file or folder is now moved to the new location.

File extension

A file extension is a suffix at the end of a filename, typically consisting of three or four characters, that indicates the type of file. The extension helps the operating system understand which program to use to open the file.

Examples of file extensions

- **.txt** – Plain Text file, for notepad
- **.jpg** or **.jpeg** – Image file (JPEG format)
- **.doc** – Microsoft Word document (97-2003)
- **.docx** – Microsoft Word document (2007 and above)
- **.xls** – Microsoft Excel (97-2003) spreadsheet
- **.xlsx** – Microsoft Excel (2007 and above) spreadsheet
- **.pdf** – Portable Document Format (Adobe Acrobat)
- **.mp3** – Audio file (MP3 format)
- **.exe** – Executable file (Windows program)
- **.html** – Web page file (HTML format)
- **.mp4** – Video file (MP4 Format)
- **.exe** – Executable file
- **.ppt** - Microsoft Power point (97-2003) presentation
- **.pptx** – Microsoft Power point (2007 and above) presentation
- **.rtf** – Rich Text File Format

File path

This is an address guide to the location of a file on a disk or network drive. The path tells the computer and user where the file is actually located.

In windows, the back slash (\) is often used to separate parts of the file path. The syntax (arrangement) for writing the file path is **path\filename.file extension**. E.g., E:\Critical data\My Notes\ICT NOTES.docx

Using the file path to identify the file name, extension, number of folders, sub-folders, and route directory

Consider the file path below:

D:\RESOURCES 2025\O LEVEL\S.4 RESOURCES\BOOKS\ ICT-SYLLABUS-compressed.pdf

File Name: ICT-SYLLABUS-compressed (This is the name without the extension.)

File Extension: .pdf

Number of Folders: There are **4 folders** in the path:

- RESOURCES 2025
- LEVEL
- S.4 RESOURCES
- BOOKS

Number of Sub-folders: Assuming "sub-folder" refers to folders nested within another:

- O LEVEL is inside RESOURCES 2025 → sub-folder 1
- S.4 RESOURCES is inside O LEVEL → sub-folder 2
- BOOKS is inside S.4 RESOURCES → sub-folder 3

So, there are **3 sub-folders** (nested levels).

Root Directory: D:\ (This is the root directory — the main drive where the path starts.)

Measurement of Computer Memory: Units of computer memory

Bit is the smallest unit of data in a computer. A bit is a binary digit represented by either 0 or 1. All data that is in the form of 0s and 1s is binary data or digital data, and is easily manipulated by the computer.

A **byte** is made up of 8 bits of information. One byte is able to store one character, e.g., x, y, 7, 9 in binary form/code. A single space between words also holds one byte.

Unit	Symbol	Bytes (Exact)	Approximation
Kilobyte	KB	1,024 Bytes	~1 Thousand Bytes
Megabyte	MB	1,048,576 Bytes	~1 Million Bytes
Gigabyte	GB	1,073,741,824 Bytes	~1 Billion Bytes
Terabyte	TB	1,099,511,627,776 Bytes	~1 Trillion Bytes
Petabyte	PB	1,125,899,906,842,624 Bytes	~1 Quadrillion Bytes

NOTE: You can get to know the amount of installed memory and other system properties on your PC when you right-click **This PC** (or **Computer**) icon and select **Properties**.

Activity: Converting from one storage measurement to another

1. A computer's hard disk has a storage capacity of 320GB. How many DVDs each of 4.7 GB can be used to provide the same storage for critical data in a business?
2. Namaganda used 12 floppy disks of 1.44 MBs each to back up her essential data for the last ten years. Her fear is that data may soon become unreadable due to outdated floppy disk technology. She now wants to transfer this data onto a flash disk of 1GB. What percentage of space shall be used on the flash disk for this data?

3. Mugisha bought a 32 GB flash drive for use to transfer data from one computer to another. The data to be transferred is 125 GB. How many times will Mugisha copy onto this flash disk to have all the data transferred to another computer?
4. A school's ICT department has received 10 external hard drives, each with a capacity of 500 GB. They plan to back up data stored on school servers totaling 3.5TB. How many of the external hard drives will be needed to store all the data? And what percentage of the hard drives will be unused?
5. A student wants to upload videos totaling 2.1 GB to a cloud service. Each of the videos is approximately 700 MB in size. How many videos can the student upload with the available space?
6. Jane has a 2 TB portable drive. She stores 400 GB of photos, 800 GB of videos, and 300 GB of documents on it. What percentage of the portable drive has she used?
7. A graphics designer stores his project files on DVDs. One of his projects requires 18 GB of storage space. If each DVD can hold 4.7 GB, how many DVDs are needed to store the entire project?
8. An IT technician is backing up files from old 2 GB USB drives. He has 20 such drives, each filled to capacity. He wants to move all the data onto a single external hard drive of 1 TB. Will all the data fit, and how much space will be left?
9. Akello is the school librarian at Greenhill Secondary School. She is responsible for backing up digital records of library inventory, students' borrowing history, and e-books collected over the years. She uses a school desktop computer with a **1 TB hard drive**, but due to increasing data volume and concerns about system failure, she plans to back up the files on external media and cloud storage.

The total data to be backed up includes: Library inventory: **10 GB**, Students' borrowing history: **3.5 GB** and E-books collection: **96.5 GB**

She has access to the following storage devices: DVDs with **4.7 GB** capacity each, USB flash drives with **32 GB** capacity and an online cloud storage account with **150 GB** of free space

Tasks:

- (a) Calculate the total size of data Akello needs to back up.
- (b) Determine how many DVDs she would need if she chose to back up all the data on DVDs alone.
- (c) If she uses the 32 GB flash drives, how many would she need to store all the data?
- (d) If she chooses to upload the data to her cloud account, what percentage of the cloud storage will be used?
- (e) Suppose Akello backs up the library inventory and student history to DVDs, and the e-books collection to the cloud. Will the cloud storage be enough? Justify your answer.

10. Mr. Ssenoga, the ICT teacher at Hillside College, is preparing digital materials for an inter-school e-learning workshop. He has organized files into three categories:

Student videos: 54 files, each approximately 250 MB

Presentation slides: 35 files, each 5 MB

Software installers: 3 files totaling 4.5 GB

He wants to carry all the materials on a 64 GB flash drive for easy transport. He also considers making DVD backups in case the flash drive fails.

Tasks:

- (a) Calculate the total size of the student videos.
- (b) Find the total size of all presentation slides.
- (c) Determine the total size of all files combined that Mr. Ssenoga wants to carry.

- (d) Will all the data fit on the 64 GB flash drive? Show your working.
- (e) If Mr. Ssenoga wants to back up the student videos on DVDs of 4.7 GB each, how many DVDs will he need?
- (f) What percentage of the flash drive's space will remain **unused** after storing all the files?

Comparing storage devices

Storage devices can be compared in terms of speed of access, portability and storage capacity, among others as summarized in the table below.

Storage Device	Access Speed	Portability	Storage Capacity
Flash disk (USB drive)	Faster access	Very portable	Moderate (4 GB – 1 TB)
Hard Disk (Internal)	Faster access	Bulky	High (500 GB – 10 TB)
Hard Disk (External)	Fast access	Bulky	High (500 GB – 10 TB)
Compact Disc (CD)	Slow	Portable	Low (Up to 700 MB)
Digital Versatile Disc (DVD)	Slow to Moderate	Portable	Low to Moderate (4.7 GB – 8.5 GB)
Magnetic tape	Very Slow	Very Bulky	Very High (Up to several TB)
Memory card (SD/TF)	Moderate	Very portable	Moderate (2 GB – 1 TB)

Burning Data to a Disc

If your computer includes CD/DVD writer, you can use it to copy files and folders to writeable discs. This process is called burning.

Windows operating system come with a preinstalled application that can burn most data files on your computer. If you have to make a disc that will play in music or video player, you should burn using a music or video program.

There are many types of optical discs, among which are:-

- (a) CD-R and DVD-R: These discs allow data to be written but the data cannot be physically erased.
- (b) CD-RW and DVD-RW: These discs allow data to be written to and erased from them.

Burning or transferring a file to an optical storage

Steps Using Windows Built-in Disc Burning Tool

Step 1: Insert the Disc

Insert a blank CD or DVD into your computer's CD/DVD drive.

Step 2: Choose What to Do

A pop-up should appear with options.

Select "Burn files to disc using File Explorer".

(If not, open File Explorer and click the CD/DVD drive under "This PC.")

Step 3: Choose Disc Format

Enter a disc title (e.g., "Backup_May2025").

Choose one of the following:

Like a USB flash drive – allows files to be added/deleted later (only works with rewritable discs).

With a CD/DVD player – finalizes the disc, suitable for media or sharing.

Click Next.

Step 4: Copy or Drag Files

Open a new window, navigate to the files you want to burn.

Drag and drop or copy and paste them into the CD/DVD window.

Step 5: Start Burning

In the CD/DVD window, click the “Drive Tools” > “Manage” tab, then click “Finish burning”.

Set the disc title (if not done before) and burning speed.

Click Next to start burning.

Step 6: Finalization

Windows will burn the files onto the disc.

Once complete, the disc may eject or show a message saying “The disc was successfully burned.”

Steps to Burn a Disc Using Ashampoo Burning Studio

Step 1: Launch Ashampoo Burning Studio

Open Ashampoo Burning Studio on your computer.

Step 2: Select a Burning Option

Choose "Data Disc" for burning files (documents, photos, videos).

If you're burning music or video files that need to be playable on standalone devices, choose "Audio CD" or "Video DVD" based on your requirements.

Step 3: Select Files to Burn

Click “Add” or drag and drop the files you want to burn into the window.

You can also browse through your folders to select files.

Step 4: Set Disc Name and Preferences

Enter a disc name (e.g., "Backup_May2025").

If desired, you can adjust the burn speed and file system (leave it as default unless you know you need to change it).

ISO9660 is common for compatibility with various devices.

UDF is better for large files and reusability.

Step 5: Insert the Disc

Insert a blank CD/DVD into your optical drive.

Ashampoo should detect the blank disc and show its capacity.

Step 6: Start the Burning Process

Once all files are selected, click the “Burn” button to start the burning process.

Ashampoo will ask for confirmation, then begin burning the files to the disc.

The software will show progress as the data is being written.

Step 7: Eject the Disc

When the process is complete, Ashampoo will notify you, and the disc will be ejected automatically, or you can eject it manually.

WORD PROCESSING 1

Word processing is the act of creating, editing, formatting, and printing text documents using a computer and a word processing application.

A **word processor** is a software program used to create, edit, format and print documents.

Examples of Word Processing Software

- Microsoft Word
- WPS Office Writer
- LibreOffice Writer
- Google Docs
- Apple Pages
- AbiWord

Documents Created Using Word Processors

- Letters
- Reports
- Memos
- Resumes
- Books
- Newsletters
- Certificates
- Flyers

Common Features of Word Processors

Feature	Description
Alignment	Arranges text: Left, Center, Right, Justify
Font	Controls text style, size, and appearance
Header & Footer	Text that appears at the top/bottom of every page in the document
WordArt	Decorative text for headings or emphasis
Symbols	Special characters (e.g., ©, ™, √) inserted into text
Screenshot	Captures and inserts an image of your screen
Images	Inserts pictures from file or online sources
Shapes	Adds arrows, rectangles, circles, etc.
Tables	Organizes data in rows and columns

Page Layout Options

Feature	Description
Margins	Space between text and page edges
Orientation	Page layout as Portrait or Landscape
Size	Page dimensions (e.g., A4, Letter)
Columns	Splits text into 2 or more columns
Indent	Moves text in from the left or right margin
Line Spacing	Vertical space between lines of text
Paragraph Spacing	Space before/after paragraphs
Borders & Shading	Adds decorative lines or background colors

Data Tools

Tool	Use
Sort	Arranges data alphabetically or numerically
Bullets & Numbering	Lists items with symbols or numbers

Formatting & Editing Text

Formatting includes changing font style, color, size, bold, italic, underline to enhance the appearance of the document.

Editing involves making corrections such as inserting, deleting, or moving text in the document.

Editing Features

These involve **modifying** or **correcting** the content of a document.

Feature	Description	Example
Cut, Copy, Paste	Move or duplicate text or objects	Cut a paragraph from page 2 and paste it on page 1
Undo/Redo	Cancel or restore the last action	Undo deletion of a table; redo inserting a picture
Delete	Remove selected text or objects	Delete a sentence or image from the document
Find and Replace	Locate specific words and optionally replace them	Replace all instances of “student” with “learner”
Select All	Highlights the entire document at once	Used before changing font style for the whole document
Insert Text	Add new text at any position	Insert a new paragraph at the top of the page
Spell Check	Identifies and corrects misspelled words	Suggests “receive” instead of “receieve”
Grammar Check	Detects grammar and sentence errors	Flags “He go to school” and suggests “He goes to school”
Thesaurus	Offers synonyms and antonyms	Replaces “happy” with “joyful”

Formatting Features

These affect the **appearance** of text and other elements in the document.

Feature	Description	Example
Font Type and Size	Change the style and size of the text	Apply “Times New Roman” in size 12
Bold, Italic, Underline	Emphasize text	Make a title bold and underlined
Font Color	Change the color of the text	Make headings appear in blue
Text Highlighting	Add background color to selected text	Highlight keywords in yellow
Text Alignment	Align text to left, right, center, or justify	Center a title, justify body paragraphs
Line and Paragraph Spacing	Adjust spacing between lines and paragraphs	Set line spacing to 1.5; add space after paragraphs
Indentation	Move text away from the margin	Indent the first line of each paragraph
Bullets and Numbering	Organize lists using symbols or numbers	Create a numbered list of steps
Borders and Shading	Add outlines and background color to text or paragraphs	Apply a shaded background to a header
WordArt	Insert stylized, decorative text	Use WordArt for a document title
Styles	Apply predefined formatting combinations	Apply “Heading 1” to section titles
Columns	Split text into multiple vertical sections	Create a two-column layout for a newsletter
Change Case	Alter capitalization of text	Change “hello” to “HELLO” or “Hello”

Proofreading Tools

Tool	Function
Find and Replace	Locates words and replaces them with other specified word
Spell Check	Highlights and corrects spelling errors

Grammar Check	Checks sentence structure and grammar
Thesaurus	Suggests synonyms and antonyms

Ribbon Tabs, Groups, and Commands

Home Tab

Group	Common Commands
Clipboard	Cut, Copy, Paste
Font	Font type, size, bold, italic, underline, color
Paragraph	Bullets, numbering, alignment, spacing, shading
Styles	Apply preset heading and text styles
Editing	Find, Replace, Select

Insert Tab

Group	Common Commands
Pages	Cover Page, Blank Page, Page Break
Tables	Insert Table
Illustrations	Pictures, Shapes, Icons, Screenshot
Add-ins	Add-ins management
Media	Online Video
Links	Hyperlink, Bookmark
Header & Footer	Insert and edit headers and footers
Text	Text Box, WordArt, Drop Cap
Symbols	Equation, Symbol

Advantages of using word processors over typewriters

Easy Editing: Word processors allow quick editing—text can be inserted, deleted, or moved without retyping the entire document, unlike with a typewriter.

Spell and Grammar Check: They automatically highlight spelling and grammar errors, helping users produce accurate documents, which typewriters cannot do.

Formatting Options: Users can change font style, size, color, and text alignment easily, whereas typewriters offer limited or no formatting choices.

Inserting Graphics: Word processors let you add images, tables, and shapes to enhance your work—typewriters can only handle plain text.

Saving and Reusing Documents: Files can be saved, reopened, and edited anytime, while typewritten pages are final and must be retyped to make changes.

Multiple Copies and Sharing: You can print or share documents electronically from a word processor. Typewriters need carbon paper for copies and rely on physical delivery.

Find and Replace: This feature helps you quickly search for words and replace them across the document—something typewriters cannot do.

Templates and Styles: Word processors offer ready-made templates and styles for common documents, saving time and ensuring consistency—unavailable in typewriters.

Disadvantages of Word processors

Dependence on Electricity and Devices: Word processors require electricity and a computer or similar device to operate. Without power or working hardware, you cannot access or edit documents.

Risk of Data Loss: Files stored digitally can be lost due to accidental deletion, hardware failure, viruses, or software crashes, especially if not backed up properly.

Cost of Equipment and Software: Using a word processor involves purchasing a computer and possibly licensed software, which can be expensive compared to a manual typewriter.

Distractions and Overreliance: The presence of other applications on computers (e.g., games, internet) may lead to distractions. Some users may also rely too heavily on spell check and ignore learning proper spelling or grammar.

Technical Skills Required: To use word processors effectively, users need basic computer skills. People unfamiliar with computers may find it difficult to navigate word processing software.

Software Compatibility Issues: Documents created in one version of a word processor may not open correctly in another, leading to formatting issues or data loss when shared between systems.

SPREADSHEET 1

Spreadsheet Software

A spreadsheet is an electronic worksheet that is used to organize data in rows and columns and perform calculations on the data.

Uses of spreadsheets

- It can be used by business men to record sales
- It can be used by business men to produce invoices
- It can be used to compile statements (e.g. Bank statements)
- Managers use them to keep track of the current payments from customers.
- It helps researchers to compile and analyze their results quickly
- Teachers can use them to create tables of figures and manipulate them quickly as required.

Advantages of using spreadsheets

- Handle a variety of applications
- Easy organizing and processing of information
- Automatic calculations
- Enables easy formatting & editing of work
- Speedy, flexible and efficient
- Can be stored and retrieved at a later time
- Reduces inventory (storage) costs
- Easy sharing over the network
- Neat work with various illustration like charts

Disadvantages of using spreadsheets

- Needs resources like computers which are expensive
- Training personnel needs resources
- Limited to power availability. It can't work without power.
- The whole work is lost in case of Virus attacks
- Data hackers can change the data with ease

- Mobility and transportation of data may be difficult especially with limited resources

Advantages of using spreadsheets over manual spreadsheets

- It is easy to make changes and corrections to data on the worksheet.
- The rest of the worksheet is recalculated whenever data on a worksheet changes.
- It is fast with the help of built-in functions and macros.
- Calculation is always accurate, provided that data and formulae entered are correct.
- It is easy to create different kinds of charts or to change chart types.
- Information on charts is updated automatically whenever related data on the worksheet changes
- Spreadsheets are much larger than manual worksheets
- They can perform mathematical, statistical and financial calculations quickly and accurately.
- Cells in electronic spreadsheets can use information from other cells
- Electronic spreadsheets can be stored and retrieved for repeated use.

Examples of spreadsheet programs

- | | |
|---------------------|---------------------------|
| • Microsoft Excel | • VisiCalc |
| • Google sheets | • OpenOffice Calc |
| • Super calc | • GS-Calc |
| • Lotus 1-2-3 | • Kingsoft spreadsheet |
| • Corel Quattro pro | • Apple works spreadsheet |

Microsoft Excel (Ms. Excel)

It is an electronic spreadsheet program that can be used for practically manipulating of data and figures.

There are two main ways you can open spreadsheet or any other program installed on a Windows computer.

- Using the Windows button (Start button)
- Using the search button

Saving a Workbook

The file that you create and save when you use the MS Excel spreadsheet is called a **Workbook**. A workbook contains several worksheets.

Remember, saving in excel is similar to saving in other applications like MS Word.

Steps to save a workbook

- Click the **file** tab
- Select **Save As** to get the Save As backstage view
- Select a location where you need to save your file for example, Desktop, Documents, and so on.
- Select a desired new location to save your work
- Type in a file name
- Click on save command

Quick Access Toolbar

Above the menu bar and on the left of the title bar is the quick access toolbar, which contains frequently used commands such as Save, Undo, Redo and others.

You can add or remove commands to the toolbar by clicking on the drop-down arrow and selecting or deselecting to add or remove commands respectively.

Ribbon Tabs

There are eight main tabs in an Excel window, each with its ribbon commands to help in manipulating information.

Clicking on one of these tabs will open the corresponding ribbon. However, the file tab is unique, with backstage view commands that affect the whole workbook/worksheet such as Save, Print and so on.

Common Terms used in Microsoft Excel spreadsheet

Cell: This is the intersection of a column and a row. Cells can contain either text, numbers or mathematical formulas.

Worksheet: A grid of columns (indicated by letters) and rows (indicated by numbers)

Spreadsheet: A computer program designed to organize numerical data in table form. Spreadsheets can manage complex calculations and collection of numbers, managed in rows and columns.

Workbook: A collection of worksheets

Row: The horizontal arrangement of cells from the left to the right of the worksheet. Each row has a number to identify it.

Column: The vertical arrangement of cells from top to bottom of the worksheet. Each column has a letter to identify it.

Cell address: A value that identifies a specific cell in a worksheet. For example, A5, is the intersection of column A and row 5.

Range: A selected group of adjacent cells, for example B3:B6.

Label: Text that is typed in a cell to identify a column or a row in a worksheet. A label therefore, can be either a column title or row title. Any entry on a worksheet that is not intended for calculation is a label.

Value: A number or text that can be used to perform calculations for example, numbers 0 to 9, currency figures like 10000, dates and so on. A value is any number, date, time, or formula that can be used in calculations.

Cell reference: A cell reference in an Excel spreadsheet is the address or name of a cell that identifies its location within the worksheet. It tells Excel where to look for data or where to place the result of a formula.

Navigating the Spreadsheet environment

Key Combination	Action / Movement
Ctrl + Home	Moves to the beginning of the worksheet
Ctrl + End	Moves to the last used cell (bottom-right of used range)
Ctrl + →	Moves to the last filled cell in the current row (right)
Ctrl + ←	Moves to the first filled cell in the current row (left)
Ctrl + ↑	Moves to the top filled cell in the current column
Ctrl + ↓	Moves to the bottom filled cell in the current column

Selecting data in MS Excel

Selecting data in Excel means highlighting one or more cells, rows, columns, or ranges so you can format, copy, move, or perform calculations on them.

Common Ways to Select Data in Excel

Action	How to Do It	Example
Select a single cell	Click on the cell	Click on B2
Select a range of cells	Click and drag across cells	A1:C5
Select entire column	Click the column letter (e.g., A, B)	Click B to select all of column B
Select entire row	Click the row number (e.g., 1, 2)	Click 3 to select all of row 3
Select multiple columns	Click and drag across column letters	Columns A to D
Select multiple rows	Click and drag across row numbers	Rows 2 to 6
Select the entire worksheet	Click the select all button (top-left corner) or press Ctrl + A	Whole worksheet
Select non-adjacent cells/ranges	Hold Ctrl and click each cell or range	Select A1:A3 and C1:C3
Select a large range quickly	Click first cell, hold Shift, click last cell	From A1 to F100

Keyboard Shortcuts for Selection

Shortcut	What It Does
Ctrl + A	Selects the whole worksheet
Shift + Arrow Keys	Expands selection one cell at a time
Ctrl + Shift + →	Selects all filled cells to the right
Ctrl + Shift + ↓	Selects all filled cells downward
Ctrl + Shift + End	Selects from current cell to last used
Ctrl + Shift + Home	Selects from current cell to A1

Inserting/Deleting Rows and Columns

To Insert	Method
Insert a row	Right-click the row number, then click Insert
	Or select a cell in the row and press Ctrl + Shift + "+"
Insert a column	Right-click the column letter, then click Insert
	Or select a cell in the column and press Ctrl + Shift + "+"
Insert multiple rows or columns	Select the number of rows/columns you want to add, then right-click and choose Insert

Note: When inserting, Excel shifts existing rows down or columns to the right.

Deleting Rows and Columns

To Delete	Method
Delete a row	Right-click the row number, then click Delete
	Or select a cell in the row and press Ctrl + "-"
Delete a column	Right-click the column letter, then click Delete
	Or select a cell in the column and press Ctrl + "-"
Delete multiple rows or columns	Select the rows/columns, then right-click and choose Delete

Filling Cells Automatically

AutoFill is a powerful feature in Excel that allows you to automatically fill cells with data, patterns, or formulas without typing them one by one.

Using the Fill Handle

The **Fill Handle** is the small square at the bottom-right corner of a selected cell or range. When you drag the Fill Handle, Excel copies or continues the pattern to adjacent cells.

Common AutoFill Uses

What You Type	What Excel Fills Automatically
1, 2	3, 4, 5... (increasing number sequence)
Monday	Tuesday, Wednesday... (days of the week)
Jan	Feb, Mar, Apr... (months)
2025-01-01	2025-01-02, 2025-01-03... (date sequence)
=A1+B1	Excel adjusts the formula when filled (e.g., =A2+B2, =A3+B3)

How to Use AutoFill

For Numbers or Dates:

- Enter the first number or date in a cell.
- Drag the Fill Handle down or across.
- Excel will either repeat or continue the sequence.

For a Custom Pattern:

- Enter two or more values to establish a pattern (e.g., 2, 4).
- Select both cells.
- Drag the Fill Handle — Excel continues the pattern (e.g., 6, 8, 10...).

AutoFill Options Menu

- Copy Cells
- Fill Series
- Fill Formatting Only
- Fill Without Formatting
- Flash Fill (for smart filling based on pattern recognition)

Flash Fill (Excel 2013 and later)

- Automatically fills data when a pattern is detected.
- Use the shortcut Ctrl + E or find it on the Data tab → Flash Fill.

Example: Type full names in one column, and first names in another. After a few entries, Flash Fill will fill the rest.

Merging Cells

Merging cells in Excel allows you to combine two or more adjacent cells into one larger cell. This is commonly used to create headings or center titles across multiple columns.

How to Merge Cells

- Select the cells you want to merge (e.g., A1:C1).
- Go to the Home tab.
- In the Alignment group, click the Merge & Center dropdown.
- Choose one of the following options:
 - Merge & Center: Merges the cells and centers the content.
 - Merge Across: Merges cells in each selected row individually.
 - Merge Cells: Merges cells without centering.
 - Unmerge Cells: Reverses the merging.

Shortcut (for Merge & Center):

Alt + H + M + C

Note:

1. Only the upper left value is left, all other values in the merged cells are deleted
2. You cannot sort or filter merged cells easily

When to Use Merging

- To create titles that span across multiple columns.
- To improve visual layout in reports or forms.

Using Clipboard: Cut, Copy and Paste

The Clipboard in Excel temporarily stores data you copy or cut, so you can paste it elsewhere. It helps you move or duplicate cell content such as values, formulas, formats, or entire cells.

Cut (Ctrl + X)

- Removes the content from the original location and prepares it to be moved.
- Border of the cut cell(s) appears dashed (moving lines).

How to Use:

Select the cell(s) → Press Ctrl + X or click **Cut** on the **Home tab** → **Clipboard group**.

Copy (Ctrl + C)

- Duplicates the content without removing it from the original cell.
- Border appears the same as with Cut (dashed lines).

How to Use:

Select the cell(s) → Press Ctrl + C or click **Copy** on the **Home tab** → **Clipboard group**.

Paste (Ctrl + V)

- Inserts the cut or copied content into a new location.

How to Use:

Click the destination cell → Press Ctrl + V or click **Paste** on the **Home tab** → **Clipboard group**.

Paste Options

After pasting, a small icon appears offering paste options like:

- **Paste Values** – pastes only the data, not formulas or formatting.
- **Paste Formulas** – pastes the formula from the original cell.
- **Paste Formatting** – applies only the formatting (like bold, color).
- **Transpose** – changes rows to columns and vice versa.

Formatting Data in Microsoft Excel

Formatting in Microsoft Excel refers to the process of changing the appearance of cells or data to make the worksheet easier to read, understand, and visually appealing. This includes adjusting things like font style, size, color, cell borders, background color, number formats, alignment, and more—without changing the actual data. In simple terms, Formatting is how you style or decorate your spreadsheet data to improve clarity and presentation.

Font Color

- Changes the color of the text in selected cells.
- Found in Home tab → Font group → Font Color (A with color bar).
- Use to highlight important data or differentiate sections.

Fill Color

- Changes the background color of selected cells.
- Found in Home tab → Font group → Fill Color (paint bucket icon).
- Useful for grouping data or improving readability.

Borders

- Adds or removes cell borders to outline cells or ranges.

- Found in Home tab → Font group → Borders (square grid icon).
- Options include: bottom, top, left, right, all borders, thick box, etc.

Orientation

- Changes the angle or direction of text inside cells.
- Found in Home tab → Alignment group → Orientation.
- Useful for rotating headings to save space or for design effects.

Commas (Thousands Separator)

- Adds commas to large numbers for easier reading (e.g., 1,000; 10,000).
- Found in Home tab → Number group → Comma Style button (,).
- Automatically formats numbers with two decimal places.

Decimal Increase and Decrease

- Adjusts the number of decimal places shown.
- Found in Home tab → Number group → Increase Decimal (→) or Decrease Decimal (←) buttons.
- Useful for controlling precision in numeric data.

Other Common Formatting Features in Excel

Feature	What It Does	Where to Find It
Font Style & Size	Change the font type (e.g., Arial, Calibri) and size	Home → Font group
Bold, Italic, Underline	Emphasize text by making it bold, italicized, or underlined	Home → Font group
Text Alignment	Align text left, center, or right within a cell	Home → Alignment group
Wrap Text	Display long text on multiple lines inside a cell	Home → Alignment group
Merge & Center	Combine selected cells into one and center the text	Home → Alignment group
Number Formats	Format numbers as currency, percentage, date, time, etc.	Home → Number group
Conditional Formatting	Automatically format cells based on rules (e.g., highlight values above 100)	Home → Styles group
Cell Styles	Apply predefined formatting sets (color, font, borders)	Home → Styles group
Row Height & Column Width	Adjust the size of rows and columns	Right-click row/column header → Row Height/Column Width
Freeze Panes	Keep rows or columns visible while scrolling	View → Window group → Freeze Panes

Using Format Painter

Format Painter is a handy tool that lets you copy the formatting from one cell (or range) and apply it to other cells without changing their content.

How to Use Format Painter

- Select the cell with the formatting you want to copy.
- Click the Format Painter button on the Home tab → Clipboard group (looks like a paintbrush).

- Your cursor changes to a paintbrush icon.
- Click or drag over the cells you want to format — the copied formatting will be applied.

Note:

- To apply the formatting to multiple areas, double-click the Format Painter button instead of single-clicking. This keeps Format Painter active until you press Esc or click the button again.
- Format Painter copies all formatting: font style, size, color, borders, fill color, number format, alignment, etc.

Formulas and Functions

Formulas are equations that perform calculations on various cells while functions are predefined formulas and are already available to perform calculations in MS Excel.

Whenever calculations are to be performed in MS Excel, always start with an equal sign (=).

Example of a formula: =B1+B2+B3+B4. A shorter way for the formula above would be the SUM function. The function can simply be type as: =SUM(B1:B4).

Note: A formula must start with “=+”

Formulas are categorized into two that is Simple formulas and Complex formulas.

Simple formulas

These are formulas with only one mathematical operators

Common Simple Formulas

Formula	Purpose	Operation	Result (if A1=10, B1=5)
=A1 + B1	Adds values in A1 and B1	Addition	15
=A1 - B1	Subtracts B1 from A1	Subtraction	5
=A1 * B1	Multiplies values	Multiplication	50
=A1 / B1	Divides A1 by B1	Division	2
=A1^2	Squares the value in A1	Exponentiation	100

Operators

Operator/Symbol	Use	Excel Formula	Meaning
+ (Plus sign)	Addition	=A1 + B1	Adds the values in A1 and B1
- (Minus sign)	Subtraction	=A1 - B1	Subtracts B1 from A1
* (Asterisk)	Multiplication	=A1 * B1	Multiplies A1 by B1
/ (Forward slash)	Division	=A1 / B1	Divides A1 by B1
^ (Caret)	Exponentiation (Power)	=A1 ^ 2	Raises A1 to the power of 2
= (Equal sign)	Equal to (comparison)	=A1 = B1	Checks if A1 is equal to B1
> (Greater than sign)	Greater than	=A1 > B1	Checks if A1 is greater than B1
< (Less than sign)	Less than	=A1 < B1	Checks if A1 is less than B1
>= (Greater than or equal to sign)	Greater than or equal to	=A1 >= B1	Checks if A1 is greater or equal

<= (	Less than or equal to	=A1 <= B1	Checks if A1 is less or equal
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<> (Not equal to)	Not equal to	=A1 <> B1	Checks if A1 is not equal to B1
& (Ampersand)	Text concatenation	=A1 & B1	Joins contents of A1 and B1

Complex Formulas

Simple formulas can be combined to create complex formulas which contain more than one mathematical operator.

For example:

- =7-2*5
- =B4+(B3-B2)/10*5

A complex formula in Excel uses two or more operators or functions in one equation. It allows you to perform multiple calculations in a single formula.

In complex formulas, the order of operations tells us which operation to calculate first. To use Excel to calculate complex formulas, the order of operations matters. Just like in regular mathematics when BODMAS or MDAS are used.

Here are the rules that determine the order:

- Parenthesis, that is operations enclosed in brackets
- Exponential calculations (to the power of e.g., 2^5, which equals 32)
- Multiplication and Division (whichever comes first)
- Addition and Subtraction (whichever comes first).

Reminder (mnemonic) that can help you recall the order such as “**P**lease **E**xercise **M**y **D**earest **A**unt **S**arah” (PEMDAS).

Example:

Suppose you enter the following formula in a cell:

=5 + 3 * 2^2

Step-by-step explanation:

- Exponents first: 2^2 equals 4.
- Then multiplication: 3 * 4 equals 12.
- Finally addition: 5 + 12 equals 17.
- So, the result Excel will display is 17.

Using Parentheses to Change the Order

If you want Excel to add first instead of multiply, you can use parentheses like this:

=(5 + 3) * 2^2

Step-by-step:

- Brackets first: 5 + 3 equals 8.
- Exponents next: 2^2 equals 4.
- Then multiplication: 8 * 4 equals 32.

So in this case, Excel gives you 32 as the result because the use of brackets changed the order of calculation.

Functions

A function is a predefined formula intended to perform calculations. Functions, like formulas, are also written with the equal sign (=) first, followed by the function name and the brackets.

Using functions is more efficient way of performing mathematical operations than using formulas. For example, if you wanted to add the values of cells C1 to C5, you would type the formula = **C1+C2+C3+C4+C5**

A shorter way of writing this formula would be using the SUM function:

=SUM(C1:C5)

There are several other functions, but our emphasis is on AVERAGE, SUM, MIN, MAX, COUNT, SMALL and LARGE.

AVERAGE

Purpose: Finds the mean (average) of a group of numbers.

Syntax: =AVERAGE(range)

Example: =AVERAGE(A1:A5) – Adds the values in cells A1 to A5 and divides by the number of values.

SUM

Purpose: Adds up numbers in a range.

Syntax: =SUM(range)

Example: =SUM(B1:B10) – Adds all the numbers from B1 to B10.

MIN

Purpose: Returns the smallest number in a range.

Syntax: =MIN(range)

Example: =MIN(C1:C6) – Returns the lowest value in that range.

MAX

Purpose: Returns the largest number in a range.

Syntax: =MAX(range)

Example: =MAX(D1:D6) – Returns the highest value in that range.

COUNT

Purpose: Counts the number of numeric entries in a range (ignores text and blanks).

Syntax: =COUNT(range)

Example: =COUNT(E1:E10) – Counts how many cells in E1 to E10 contain numbers.

SMALL

Purpose: Returns the k-th smallest number in a list (e.g., 2nd smallest, 3rd smallest).

Syntax: =SMALL(range, k)

Example: =SMALL(F1:F5, 2) – Returns the second smallest number in F1 to F5.

LARGE

Purpose: Returns the k-th largest number in a list (e.g., 1st largest, 3rd largest).

Syntax: =LARGE(range, k)

Example: =LARGE(G1:G5, 1) – Returns the largest number (1st largest) in G1 to G5.

Sorting Data

Sorting in Excel allows you to arrange your data in a specific order so it's easier to read, analyze, or present. You can sort numbers, text, or dates in ascending or descending order.

Types of Sorting

Ascending Order

- Numbers: From smallest to largest (e.g., 1 → 100)
- Text: From A to Z
- Dates: From oldest to newest

Descending Order

- Numbers: From largest to smallest (e.g., 100 → 1)
- Text: From Z to A
- Dates: From newest to oldest

Steps to Sort Data

- Select the range of cells you want to sort (including headers if available).
- Go to the Home tab or Data tab on the ribbon.
- Click on Sort & Filter.
- Choose:
 - Sort A to Z for ascending
 - Sort Z to A for descending
 - Or click Custom Sort for more options

Custom Sort Options

- Sort by specific columns (e.g., sort by "Name", then by "Age")
- Sort by cell color, font color, or cell icon
- Add multiple sorting levels (e.g., sort by Class, then by Score)

Using Charts to Represent Data in MS Excel

A chart is an object used to communicate numerical data in a graphical format. Many people use charts to summarize numerical data and display it in an easy to interpret layout or predict current and future trends in data. Charts in Excel help you visually represent data so that patterns, trends, and comparisons are easier to understand. Excel offers a variety of chart types to suit different types of data.

Why Use Charts?

- To make data easier to interpret
- To highlight comparisons or changes over time
- To support presentations and reports with visual evidence

Types of charts

Column Chart

A column chart displays data as vertical bars (columns). Each column represents a value for a specific category.

It is best for comparing individual values across categories and showing changes in values over time (when time is grouped by category).

Example Use:

- Comparing marks of students in different subjects.
- Sales per product in a single month.

Bar Chart

A bar chart is similar to a column chart, but the bars are horizontal instead of vertical.

It is best for comparing values across categories when the category names are long or many and displaying survey results or rankings.

Example Use:

- Comparing population sizes of different cities.
- Ranking students by total scores.

Pie Chart

A pie chart shows data as slices of a circle, where each slice represents a portion of a whole.

It is best for displaying proportions or percentages and showing how one category compares to the total.

Example Use:

- Showing percentage of expenses in a budget.
- Displaying student attendance as a percentage.

Line Chart

A line chart displays data points connected by lines, usually to show trends over time.

It is best for tracking changes and trends over time (days, months, years) and observing rise or fall in values.

Example Use:

- Plotting monthly temperatures.
- Showing student performance across terms.

Steps to Create a Chart in Excel

- Enter your data into a worksheet (include labels for clarity).
- Select the data range you want to chart (include headings).
- Go to the Insert tab on the ribbon.
- Choose a chart type from the Charts group (e.g., Column, Line, Pie).
- The chart will appear on your worksheet. You can move and resize it.

Customizing a Chart

Use the Chart Tools to:

- Add or edit titles and labels
- Change colors, fonts, and styles
- Add data labels or legends
- Apply chart layouts and designs

Tips

- **Always** label your chart clearly (title, axis labels, and legend).
- Choose the chart type that best fits your data.
- Use data labels to make charts more informative.

Chart elements and their uses or meanings

Chart Title

- Describes what the chart is about (purpose of the chart).
- Gives context so viewers understand what data the chart represents.

Axis Titles

- Labels for the horizontal (X-axis) and vertical (Y-axis) axes.
- Explains what the numbers or categories on each axis represent.

Legend

- Identifies what each color or pattern in the chart means.
- Helps distinguish between different data series or categories.

Data Series

- The actual data points or bars/lines/pie slices that represent your values.
- Visual representation of the numbers from your data.

Data Labels

- Shows the exact value of each data point directly on the chart.
- Makes it easier to see specific values without referring to the table.

Gridlines

- Horizontal and/or vertical lines that run across the chart.
- Help read and compare values more easily by providing a reference.

Plot Area

- The area where the data series is plotted.
- The main part of the chart that displays the data visually.

Chart Area

- The entire space of the chart, including titles, labels, and plot area.
- The whole chart object within the worksheet.

Axis

- Lines along which data points are plotted (X-axis and Y-axis).
- Define the scale and categories of data.

Uses of spreadsheets in life

- Spreadsheets help individuals keep track of income and expenses, categorize spending, and calculate monthly savings or deficits, making it easier to manage finances effectively.
- Teachers and students use spreadsheets to enter and calculate marks across subjects, generate totals, averages, and monitor performance throughout the term or year.
- Spreadsheets can be used to create structured timetables showing days, periods, subjects, and teachers, which makes it easier to plan and organize academic schedules.
- People and businesses use spreadsheets to log daily, weekly, or monthly expenditures, which helps in monitoring spending patterns and controlling unnecessary costs.
- Data collected from surveys can be entered into spreadsheets and analyzed using functions and charts to identify trends, preferences, or opinions among respondents.
- Spreadsheets assist in organizing event details like guest lists, budgets, schedules, and tasks to ensure everything is properly arranged and managed.
- Businesses and individuals can use spreadsheets to list items, their quantities, prices, and locations, which helps in maintaining accurate and updated stock records.
- Spreadsheets can perform calculations to determine loan repayments, interest earned, or mortgage schedules using formulas, saving time and improving accuracy.
- Employers use spreadsheets to calculate salaries, deductions, bonuses, and taxes for employees, generating summaries and payment details quickly.
- People use spreadsheets to prepare and organize shopping lists, sometimes with prices and quantities, helping them budget and avoid forgetting items.
- Spreadsheets can be used to log exercise routines, track weight loss or gain, and monitor nutrition, helping individuals stay on top of their health goals.
- Travelers can use spreadsheets to list destinations, budgets, packing items, and schedules, ensuring nothing important is missed during trips.
- Spreadsheets enable the creation of data-based reports with summaries, totals, and charts, useful for presentations in school, work, or meetings.
- Spreadsheets help predict future outcomes like sales, profits, or performance by using historical data and applying functions or trendlines for analysis.

Headers and Footers in a worksheet

A header is a piece of data placed in the top margin of every page.

A footer is a piece of data placed in the bottom margin of every page in the document.

Headers are commonly used for page titles and organization names and footers are suitable for page numbers, author's name, footnotes and dates/time.

Inserting header/footer in a worksheet

Step 1: Open Your Excel Worksheet

- Make sure the worksheet where you want to insert the header or footer is open.

Step 2: Go to the "Insert" Tab

- Click the Insert tab on the ribbon at the top of the screen.

Step 3: Click on "Text" Group

- In the Insert tab, locate the Text group.
- Click Header & Footer.

Step 4: Worksheet Switches to Page Layout View

- Excel will automatically change the view to Page Layout View.
- You will see Header sections (Left, Center, Right) at the top of the page.
- Scroll to the bottom to see the Footer sections.

Step 5: Type Your Header or Footer

- Click into the desired section (Left, Center, or Right).
- Type your text, such as the title of the worksheet, date, or page number.

Step 6: Use Header & Footer Tools

- When you're in header/footer mode, the Header & Footer Tools Design tab appears.
- Use this tab to:
 - Insert page numbers
 - Insert current date or time
 - Insert file name or sheet name
 - Add a picture (e.g., school logo)

Step 7: Exit the Header/Footer Area

- Click anywhere outside the header or footer area to return to normal view.
- Or switch back to Normal View via the View tab > Normal.

Printing in MS Excel

Printing in Excel allows you to produce a hard copy of your worksheet, ensuring that your data is presented clearly and professionally.

Steps to Print in Excel

- Open the Excel file you want to print.
- Click the File tab in the top-left corner of the screen.
- Select Print from the list. This opens the Print Preview window where you can see how your document will look on paper.
Select the printer you want to use from the list of available printers.
- Set Print Options
 - ✓ Number of copies: Choose how many copies you want.
 - ✓ Print Active Sheets: Print the current sheet or select another option like "Print Entire Workbook" or "Print Selection".
 - ✓ Pages: Choose to print all pages or a specific range (e.g., pages 1–2).
- Adjust Page Setup (Optional)
 - ✓ Change orientation (Portrait or Landscape)
 - ✓ Set paper size
 - ✓ Adjust margins
 - ✓ Fit the worksheet to one page wide or tall
- Click the Print button to start printing your document.

ELECTRONIC PRESENTATION

An electronic presentation has a sequence of text, graphics, animations and sometimes audio and video to accompany speech or public presentation.

Presentation Software

Presentation software is a type of application used to create visual aids, usually in the form of slides, to support verbal communication during meetings, lectures, or speeches.

Examples of Presentation Software

- Microsoft PowerPoint
- Google Slides
- Apple Keynote
- LibreOffice Impress
- Canva (Presentation tool)
- Harvard Graphics
- OpenOffice Impress
- Prezi
- Kingsoft presentation
- Corel presentation

Microsoft PowerPoint

Microsoft PowerPoint is a presentation software developed by Microsoft, widely used for creating slideshows that include text, images, charts, audio, video, and animations to support spoken presentations.

Microsoft PowerPoint has evolved through various versions over the years, each introducing new features and improvements. Early versions included PowerPoint 97 and 2000, which provided basic slide creation and formatting tools. PowerPoint 2003 introduced improved design and multimedia support. PowerPoint 2007 brought the Ribbon interface and new file formats (.pptx). PowerPoint 2010 enhanced video and image editing features, while PowerPoint 2013 added cloud integration and widescreen slide support. PowerPoint 2016 introduced real-time collaboration and design ideas. PowerPoint 2019 refined these features and improved 3D and Zoom tools. The latest version, PowerPoint from Microsoft 365 (formerly Office 365), offers continuous updates, AI-powered design suggestions, online sharing, and improved collaboration features.

Opening MS PowerPoint

To open Microsoft PowerPoint on your computer, follow these steps:

Method 1: Using the Start Menu (Windows)

- Click the Start button (Windows icon) at the bottom-left corner of your screen.
- Scroll through the list of programs or type PowerPoint in the search bar.
- Click on Microsoft PowerPoint from the results to open it.

Method 2: Using a Desktop Shortcut

- If there is a PowerPoint shortcut icon on your desktop, double-click it to open the application.

The MS PowerPoint Interface

- Title Bar: Displays the name of the presentation and the application.
- Menu bar: The strip at the top containing tabs like Home, Insert, Design, Transitions, Animations, Slide Show, Review, and View. Each tab has groups of related commands and tools.
- Ribbon: The MS PowerPoint ribbon is used in the same way as the one in MS Word. There are several tabs on the ribbon e.g., Home, Insert and others. By clicking on each tab displays different sets of tools on the ribbon.
- Quick Access Toolbar: Located above the Ribbon, it provides shortcuts to frequently used commands like Save, Undo, and Redo.

- Slides Pane: Usually on the left side, it shows thumbnails of all slides in your presentation for easy navigation.
- Slide Workspace: The large area in the center where you design and edit your current slide.
- Notes Pane: Below the Slide Workspace, where you can add speaker notes that won't show on the slide but can be used during presentations.
- Status Bar: At the bottom, it shows slide number, design theme, and view options like Normal, Slide Sorter, or Slide Show.
- Zoom Controls: Located on the status bar, allowing you to zoom in or out on the slide.

Working with MS PowerPoint

Inserting a new slide

To add a new slide to your presentation, follow these steps:

Method 1: Using the Ribbon

- Go to the Home tab on the Ribbon.
- Click the New Slide button in the Slides group.
- A new slide with the default layout will be inserted after the currently selected slide.
- To choose a different layout, click the small arrow below the New Slide button to see layout options, then select one.

Method 2: Using Keyboard Shortcut

- Press Ctrl + M (Windows) to quickly insert a new slide with the default layout.

Method 3: Using the Slides Pane

- Right-click in the Slides Pane (the thumbnail list on the left).
- Select New Slide from the context menu.

Adding Content to Slides

After inserting a slide, you can add different types of content to make your presentation informative and engaging:

Text

- Click inside a text placeholder or text box on the slide.
- Start typing your content like titles, bullet points, or paragraphs.
- Use the Home tab to format text (font, size, color, alignment).

Images

- Go to the Insert tab.
- Click Pictures to add images from your computer or Online Pictures to search the web.
- Select and insert the image, then resize or move it as needed.

Shapes

- On the Insert tab, click Shapes.
- Choose a shape like rectangle, circle, arrow, etc., and draw it on the slide.
- Customize color, outline, and effects using the Drawing Tools.

Charts

- Click Insert > Chart to add bar, line, pie, or other charts.
- Enter your data in the spreadsheet that appears to update the chart.

Videos and Audio

- Under Insert, select Video or Audio.
- Choose a file from your computer or online to embed it into the slide.

SmartArt

- Use Insert > SmartArt to add diagrams like processes, hierarchies, or cycles.

Basic guidelines for creating presentation

Here are some basic guidelines for creating an effective presentation:

- Keep slides simple and uncluttered; focus on key points only.
- Use large, readable fonts and limit the number of different font styles.
- Choose high-contrast colors for text and background to ensure readability.
- Use visuals like images, charts, and diagrams to support your message.
- Avoid overcrowding slides with too much text or too many graphics.
- Maintain consistency in design, font, and colors throughout the presentation.
- Use bullet points or short phrases instead of long paragraphs.
- Include slide titles that clearly describe the content.
- Limit animations and transitions to avoid distractions.
- Practice your speech to align well with the slide flow and timing.
- Always check spelling and grammar before presenting.

Formatting Content in Slides

To change the appearance of content on your slides, you must select the slide and, or its contents. Then you format your text using an appropriate command on the **Home tab**.

You can also change the appearance of slides by using or applying slide design features.

To format content requires one to first select the text or object and then click the appropriate command to format it. Most formatting command are found on the Home tan in the Font group and also use the Paragraph group of commands.

Some of the formatting commands include;

- Creating/modifying bulleted and numbered lists
- Change case
- Applying shadow to text especially headings
- Applying color to text
- Aligning text within textbox
- Spacing between paragraphs and lines or points

Animations in PowerPoint

Animation effects are features that are applied to text and objects to give them motion in a slide show presentation. You can apply animation effects to takes and images appropriately to emphasize points, generate excitement and also to help the audience understand your message. However, too much amination can be distracting

Transition effects

A transition is a feature used to control how slides move, one after the other, in a slide show presentation. When you add a transition effect to a slide, the effect takes place between the current slide and the previous slide.

Using hyperlinks in presentations

A hyperlink is a connection from one slide to another slide in the same presentation or to a slide in another presentation, an e-mail address, a web-page, or a file. You can create hyperlink from text or from an object such as a figure, graph, shape or WordArt.

Inserting Shapes in a presentation

A shape is a sketch of an object. There are many shapes you can add in a presentation to enhance it.

Examples of Shapes in PowerPoint

- Rectangles: Used for boxes, containers, or backgrounds.
- Circles and Ellipses: Used for icons, logos, or highlighting important information.
- Arrows: Used to show direction, flow, or connections.
- Lines: Used to separate sections, create borders, or add visual interest.
- Callouts: Used to highlight important information or add captions.
- Stars and Banners: Used to add visual interest or highlight key points.

How to Insert Shapes

- Click on the "Insert" tab in the PowerPoint ribbon.
- Click on the "Shapes" button in the "Illustrations" group.
- Select the shape you want to insert from the dropdown menu.
- Click and drag on the slide to draw the shape.
- Use the "Format" tab to adjust the shape's fill, outline, and effects.

Inserting images in a presentation

Adding images in your presentation makes it more interesting and attractive.

To insert images in a PowerPoint presentation:

- Click on the "Insert" tab in the PowerPoint ribbon.
- Click on the "Pictures" button in the "Images" group.
- Select the source of the image:
 - o This Device: Insert an image from your computer.
 - o Stock Images: Use pre-installed stock images.
 - o Online Images: Search for images online.
- Choose the image you want to insert.
- Click "Insert" to add the image to your slide.

Additional Options

- Drag and drop images: You can also drag and drop images directly into PowerPoint.
- Use image placeholders: Insert image placeholders to reserve space for images.

Inserting Tables in a presentation

Tables can be used in a presentation to show numerical data such as price lists, budgets or performance of business. In MS PowerPoint, you can create table or insert table from another application such as MS Word.

To insert tables in a PowerPoint presentation:

- Click on the "Insert" tab in the PowerPoint ribbon.
- Click on the "Table" button in the "Tables" group.
- Select the number of rows and columns you want for your table.
- Click to insert the table into your slide.

Alternative Method

- Click on the "Table" button and select "Insert Table" to specify the number of rows and columns.

Inserting Charts in presentation

Charts are useful in representing numerical data graphically. An existing chart/graph can be copied from another application or you can create a new chart using PowerPoint.

To insert charts in a PowerPoint presentation:

- Click on the "Insert" tab in the PowerPoint ribbon.
- Click on the "Chart" button in the "Illustrations" group.
- Select the type of chart you want (e.g., column, line, pie).
- Enter your data in the Excel spreadsheet that opens.
- Customize the chart. Adjust chart title, labels, and formatting as needed.

Inserting SmartArt in a presentation

SmartArt is a feature in PowerPoint that allows you to create visually appealing diagrams and illustrations such as Venn diagrams, pyramids and organizational charts.

To insert SmartArt in a presentation:

- Click on the "Insert" tab in the PowerPoint ribbon.
- Click on the "SmartArt" button in the "Illustrations" group.
- Select a category (e.g., List, Process, Cycle) and choose a SmartArt graphic.
- Type your text into the SmartArt graphic.
- Customize the SmartArt. Adjust colors, styles, and layout as needed.

Inserting Media clips

A media clip can be an audio or a video that is used to supplement information in a presentation. When in a presentation, media clips can enhance the audience's ability to pick up the message in the presentation.

To insert media clips in PowerPoint:

- Click on the "Insert" tab in the PowerPoint ribbon.
- Choose the type of media clip you want to insert:
 - o Video: Insert a video file from your computer, online sources, or stock videos.
 - o Audio: Insert an audio file from your computer or online sources.
- Choose the location of your media file:
 - o This Device: Insert a file from your computer.
 - o Online Video: Search for videos online.
 - o Stock Videos: Use pre-installed stock videos.
- Click **"Insert"** to insert the media clip into your slide.

Headers and Footers

Headers and footers in PowerPoint help provide important contextual information (such as dates, slide numbers, or custom text) consistently across slides or handouts. Unlike Word, PowerPoint has limited header functionality on slides, but you can still add useful footer content.

How to Insert Headers and Footers on Slides

- Go to the Insert tab on the ribbon.
- Click Header & Footer in the Text group.
- In the Header and Footer dialog box:
 - o Under the Slide tab:
 - Check Date and time to include current or fixed date.
 - Check Slide number to show numbering.
 - Check Footer and type custom text (e.g., "Confidential").
 - Check Don't show on title slide if needed.
- Click Apply to All to apply to every slide or Apply for the current slide only.

PowerPoint Presentation Views

PowerPoint has a number of views which can be used to display a presentation. By default, when you open PowerPoint, it opens in the Normal View that mainly displays the Slide pane and outline pane. You can add speaker's notes at the bottom. This view is used to create/edit slides.

Normal View

Normal View is the default and most commonly used view in PowerPoint, designed for creating and editing slides. It displays a slide pane in the center, a thumbnail list of slides on the left, and an optional notes section at the bottom. This view allows you to work on the content of individual slides, insert images, text, and animations, and organize slide elements easily.

Slide Sorter View

Slide Sorter View presents all the slides in your presentation as thumbnails on a single screen. This view is ideal for quickly rearranging slide order, applying transitions between slides, and gaining an overall sense of the flow and structure of the presentation. It is particularly useful when organizing or refining the layout of a large number of slides.

Reading View

Reading View displays the presentation in full-screen mode but still allows access to the Windows taskbar. It is meant for reviewing your slides as they will appear during a presentation, without switching to full Slide Show mode. This view is especially useful when you want to preview your presentation while continuing to use other applications.

Slide Show View

Slide Show View is used to present your slides to an audience. It runs the presentation in full screen, hiding all editing tools and allowing you to navigate through the slides using the keyboard, mouse, or a remote. This view is essential for delivering a polished, distraction-free presentation during meetings, lectures, or events.

Notes Page View

Notes Page View shows one slide at a time along with its accompanying speaker notes. This view is helpful for editing or printing notes that can assist you during your presentation. It is often used by presenters to prepare detailed talking points that are not visible to the audience during the slide show.

Outline View

Outline View focuses solely on the text content of your slides, showing the titles and bullet points in a collapsible list format. It helps you organize the structure and flow of your presentation quickly without being distracted by graphics or layout. This view is useful for content development and planning before finalizing slide designs.

Master Views

Master Views, including Slide Master, Handout Master, and Notes Master, allow you to define global design elements and layouts that apply to the entire presentation. These views let you customize themes, fonts, placeholders, and footers to ensure consistency across all slides, handouts, or notes. They are ideal for setting a professional and uniform style throughout the presentation.

Slide Master in PowerPoint

Slide Master is a special feature in PowerPoint that allows you to control the overall design and layout of your presentation from a single location. It provides a master slide template where you can define default settings like fonts, colors, background styles, placeholders, and logos, which will automatically apply to all slides that follow that master layout.

Purpose of Slide Master

- Ensures consistency across all slides.
- Saves time by letting you apply changes (e.g., font or logo) to multiple slides at once.
- Helps maintain a professional and uniform look in your presentation.

Printing a Presentation in PowerPoint

Printing a presentation in Microsoft PowerPoint allows you to create hard copies for audience handouts, presenter notes, or archival purposes. PowerPoint offers several print layout options to suit different needs, such as full slides, handouts with multiple slides per page, or slides with notes.

Printing Slides

- On the **File** tab, choose Print or press Ctrl + P; the print back stage view (dialog box) displays.
- Type the number of copies to print
- Select the **printer** to use, if it is not the default printer
- Select slides to print.
 - o **Print All slides** (Prints entire presentation)
 - o **Current slide** (prints the selected slide)
 - o **Custom range** (prints specific slides typed in the slides box)
- Select what to print:
 - o **Full page slides**: This prints a full slide per page
 - o **Notes page**: This prints one slide with its presentation notes on each page
 - o **Outline**: This prints an outline of the presentation
 - o **Handouts**: This prints as many slides as you set on a page up to a maximum of 9 slides.
- Select collated to print a full copy of the presentation before another copy is printed.
- Finally click on the **Print** button.

INFORMATION ACCESS AND SHARING

Information Access

Information Access refers to the ability to identify, retrieve and use information effectively. Traditionally information has been disseminated in a variety of formats that have even been widely accessible, often through public libraries. Many people also relied on other people and media for information.

The advances in computer technology have revolutionized information access, making vast stores of information about business, education, health, government, and entertainment easily accessible on the internet.

However, even without internet, one can access information though not easily. Information can be accessed either online or offline.

In computer technology, online means a state of being connected to internet whereas offline indicates the state of being disconnected.

Online access is preferred because its information can be accessed anywhere and anytime provided there is an internet connection. Offline sources can become outdated. Online sources require internet access which may be expensive

Offline Information Access

Nowadays, there are computer programs which can be installed to allow users to access a vast amount of information without necessarily having an internet connection. Some of these platforms include Britannica and Encarta.

Information sharing

Information sharing involves the exchange of data, knowledge, or insights between individuals, teams, departments, organizations, or even across different sectors.

The primary goal of information sharing is to facilitate collaboration, improve decision-making, enhance problem-solving, and promote a culture of knowledge sharing and continuous improvement.

Information sharing can occur through various methods, including direct communication, electronic platforms, databases, and shared documents.

Importance of Information Sharing

Sharing information amongst people in society helps people to make informed decisions. Sharing information helps to enhance innovation and growth because people can research about what others have done before, the challenges faced and possible solutions.

It helps to improve service delivery to customers and hence retain them.

It helps to improve the welfare of the society. For example, when the Ministry of Health in Uganda shares information on how to avoid Coronavirus, Cholera and HIV/AIDS.

Online Information Access

This refers to searching for information on the internet. Internet is a network of worldwide computer networks connected together to communicate and share resources. The primary purpose of the internet is to facilitate access and sharing of information.

How the Internet Works?

To connect a device to internet, you need an Internet Service Provider (ISP) such as MTN Uganda. An ISP is a company that provides internet services to individuals and companies. Your computer should also have a web browsing software such as Internet Explorer or Microsoft Edge or Google Chrome.

Each computer that connects to the internet is assigned a unique address by the ISP. This address is often called an IP address (Internet Protocol address). The IP address identifies each computer uniquely on the internet.

The History of Internet

History of the Internet: In 1969 the U.S. Department of Defense commissioned the Advanced Research Projects Agency Network (ARPANET). The purpose was to provide communication links between supercomputers located at various regional sites (Universities and defense bases) within the United States.

It is this ARPANET that evolved into the Internet after computer networks were connected to it in different parts of the world. By 1992, more than 1 million hosts existed on the Internet.

Requirements for Connecting to the Internet:

- ✓ **Computer** - two or more
- ✓ **Network Adaptor (Network Interface Card):** The NIC may be based on Ethernet or wireless technologies.
- ✓ **An Internet Service Provider (ISP):** the company that takes care of the technical aspects of connecting to the internet. ISPs available in Uganda include Mobile telephone companies like Lyca Mobile, MTN, Airtel, UTL, Smile etc.
- ✓ **Modem:** This is a device that converts analog telephone signals into digital computer information and vice-versa. Some computers have internal / inbuilt modems. These include wireless modem, Integrated Services Digital Network (ISDN), Digital Subscriber Line (DSL), Cable modems.
- ✓ **Required Software:** programs necessary to use the internet services such as web browsers and Email Clients.

Factors that affect the speed of an internet connection

Computer processor speed and power: The lower the computer speed, the lower the speed of the internet connection will be. A computer whose processor is Pentium IV has a slower internet connection than one with a Core i5 processor.

Heavy traffic on the network: The greater the number of computers connected to the same network, the slower the speed of the internet connection will be and the reverse is very true.

Modem speed: Modems, WIFI dongles and MIFIs always have their maximum speeds specified by the manufacturer. Therefore, the lower the speed of the device, the slower the internet connection will be and the reverse is true.

Hardware problems: A poor network card or signal receiver will lead to a slow internet connection speed and vice versa.

Software problems: Using older versions of software like browsers leads to a slow internet connection and the reverse is true.

Cookies: Cookies refers to the information like passwords collected and stored by a browser whenever the user visits the internet. As the cookies become many, they may compromise the functionality of the browser leading to a slow internet connection.

Technical problems: Slow internet connection can be due to technical issues cause by the ISP. These include; maintenance works and system updates being done by the ISP.

Natural conditions: Rough conditions like storms, heavy down-pours, among others lead to a slow internet connection speed and the reverse is true. This is because they sabotage the correct

transmission of wave signals.

Positioning of Routers and signal receivers: Poor positioning of devices (routers, modems,

WIFI, MIFI) leads to a slow internet connection and vice versa. This is because they (devices) find difficulties in receiving and sending signals.

Implications of the Internet (advantages/Disadvantages)

Advantages

- (a) Offers personnel connectivity for friends and financiers through E- mails, E–collaboration and chat – rooms, online telephone calls, voice mails, video phones, etc
 - (b) It is a tool for entertainment through on–line games, clubs and interactive games, online chats, refreshing video clips (movies) and animals” zoos.
 - (c) It is a source of important electronic programs/applications through downloading of freeware, shareware and other programs.
 - (d) Offer travel services like E-Booking and reservations for aviation firms and hotels, to business travelers and holidaymakers.
 - (e) Is a tool for financial management offering answer to financial management challenges of where, when and how to invest especially in financial securities.
- Success belongs to the most persevering Page 81
- (f) Offers education and research facilities for either academic or knowledge enrichment. E- Publishing and E- Learning have greatly improved distance learning. Hence WWW being a body of knowledge. The common wall-less classrooms
 - (g) Multimedia: offers Audio and video services.
 - (h) Offers Banking services for online Banking institutions, and electronic funds transfers (EFT). A case for western union, Digi Cash, Cyber Cash, Millicent, ATMs, and credit card facilities are in this category.
 - (i) Offers news updates (information Retrieval) for politics, social events, climate and weather, and sports. These usually in form of live radio, live TV, online newspapers and magazines, educational short films, etc.
 - (j) On – line employment for job seekers through online employment agencies, and firms advertising jobs online.
 - (k) E–Medicine where diseases diagnosis and drug prescriptions are done on-line
 - (l) E-Courtships and marriages, etc
 - (m) Telecommuting. Through internet connectivity and interactivity one can now work from home and communicate with the office through networked computers and telephone facilities.
 - (n) A tool for transacting business through E-commerce or E- Business where products can be marketed

Disadvantages of using the Internet

- Computer viruses can be spread over the internet.
- The Internet provides access to unsuitable material such as Pornography, the biggest threat to healthy mental life.
- Theft of Personal information: If you use the Internet, you may be facing grave danger as your personal information such as name, address, credit card number etc. can be accessed by hackers or thieves.
- Spamming: Spamming refers to sending unwanted emails in bulk, which provide no purpose and needlessly obstruct the entire system.

- Some people get addicted to the internet, causing problems with their social interactions with friends and loved ones.
- The initial cost of connecting to the internet is high. e.g., buying computers.
- Many people are computer illiterate and so cannot use the internet, hence miss.
- There is a lot of wrong information on the internet. Anyone can post anything, and much of it is garbage/ inaccurate.
- Information interceptions -Information carried on ICTs channel especially on internet has no guaranteed security coz hackers can gain access to secret and important information- credit card PIN during online payments.
- Corn men who practice phishing & pharming on the internet who end host websites claim to have genuine business & end up robbing people.
- Addictions especially computer games plus surfing by young people leads to wastage of valuable time.
- Some of the sites on the internet require passwords to have access to information that you want.
- Internet scams where people are conned property and lose life at times. The most recent true story is of a European dater who was abducted in West Africa and made to pay a ransom for his release while looking.

Internet Service Provider

An internet service provider (ISP) is a company that supplies connections to the internet usually for a monthly fee. Users may connect to their ISP through a LAN or through dial-up-access. A dial-up access is a slow speed technology.

Examples ISPs in Uganda

- Mobile Telecommunications Network (MTN)
- Airtel
- Africell
- Smile
- Vodafone
- Infocom
- Uganda Telecommunications Network. (UTL)
- Lyca Mobile

Uniform Resource Locator

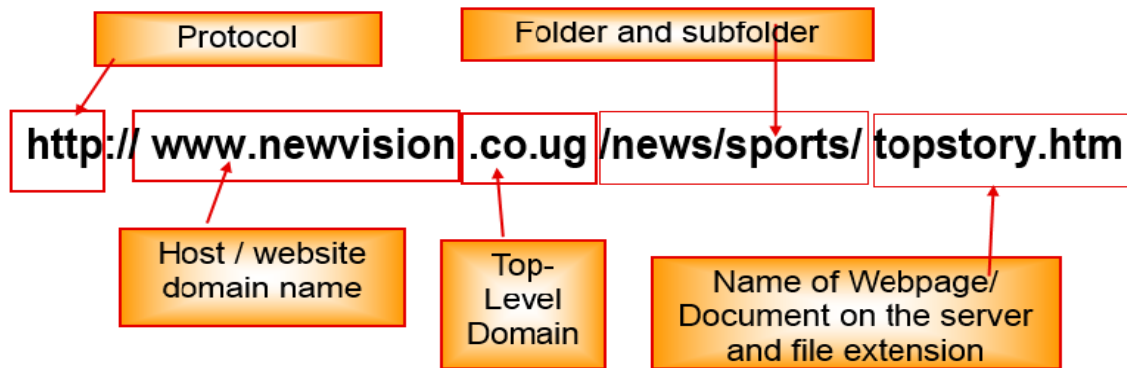
This is a reference to a web resource that specifies its location on a computer network and a mechanism for retrieving it.

A Uniform Resource Locator is a unique identifier that locates a resource on the internet. It is also called a web address. To identify a Web pages exact location, Web browsers rely on a Uniform Resource Locator (URL).

This is the unique address of a web page. The URL tells the browser where to locate the names and sometimes the path to a specific web page or location on a web page

To identify a Webpages exact location, Web browsers rely on a Uniform Resource Locator (URL).

Structure of a Uniform Resource Locator



URL Example

http://www.sc.edu/beaufort/library/pages/bones/lesson1.html

Here's what it all means:

"http" means hypertext transfer protocol and refers to the rules used to transfer and deal with information

"www" stands for World Wide Web and is the general name for the host server that supports text, graphics, sound files, etc. (It is not an essential part of the address, and some sites choose not to use it)

Parts of the URL

"sc" is the second-level domain name and usually designates the server's location, in this case, the University of South Carolina

"edu" is the top-level domain name (see below)

"beaufort" is the directory name

"library" is the sub-directory name

"pages" and **"bones"** are the folder and sub-folder names

"lesson1" is the file name

"html" is the file type extension and, in this case, stands for "hypertext mark-up language" (that's the language the computer reads).

Web Browsers

A **web browser** is a software application that allows users to access, retrieve, and display content from the World Wide Web. It interprets and renders web pages written in HTML, enabling users to interact with websites.

Examples of Web Browsers

Web Browser	Developer	Features
Google Chrome	Google	Fast, secure, supports extensions, widely used.
Mozilla Firefox	Mozilla Foundation	Open-source, privacy-focused, customizable.
Microsoft Edge	Microsoft	Integrated with Windows, efficient, secure.
Apple Safari	Apple	Optimized for macOS & iOS, energy-efficient.
Opera	Opera Software	Built-in VPN, ad blocker, and speed dial.
Brave	Brave Software	Privacy-focused, blocks ads and trackers.
Tor Browser	Tor Project	Designed for anonymous browsing.
Vivaldi	Vivaldi Technologies	Highly customizable, advanced tab management.

Search Engine

A search engine is a program or tools that finds information from the World Wide Web containing keywords specified by users. The information is searched from databases and internet sites. Search engine is usually integrated in a web browser.

To search information from the web, the user must first connect to the internet using a web browser and then type keywords to be searched in a search engine such as Google (www.google.com).

Examples of search engines

- | | |
|--------------|--------------|
| ✓ Google | ✓ Excite |
| ✓ Dogpile | Lycos |
| Yahoo | ✓ Hot Bot |
| ✓ Alta vista | MSN |
| Web crawler | ✓ Ask Jeeves |

Effective Internet Searching

The challenge is to ask your question the right way, so that you don't end up overwhelmed with too many search results, underwhelmed with too few, or simply unable to locate the material that you need.

Keywords: Search engines don't read sentences the way people do: instead, they look for the key words in your query in the websites they search. Common words are ignored (that, to, which, a, the, ...).

- Use "**quotation marks**" to search as a phrase and keep the words linked together.
- **+** and **-** can be used to include or exclude a word
- Boolean Syntax: Enter words and connect with Boolean Operators: AND, OR, NOT
 - **AND** will include sites where both words are found. Uses: joining different topics eg (ie. "Global warming" AND California).
 - **OR** - requires at least one of the terms is found. Uses: join similar or synonymous topics (ie. "Global warming" OR "greenhouse effect")
 - **NOT** – searches for the first term and excludes sites that have the second term
Uses: join similar or synonymous topics (ie. Washington NOT school)

Technologies Used to Share Information on the Internet

Using emails to share information

Email is an acronym of electronic mail.

Email communications refers to the transmission of messages via computer networks such as; a local area network, WANs, or internet.

The email can be simple text, or include an attachment such as a word processing document, PDF or graphic.

Email software (Commonly known as email client) is a computer program used to access and manage a user 's email account. It may be web based or not. Popular locally installed email clients include **Microsoft Outlook, Pegasus Mail, KMail, Evolution and Apple Mail**. Popular web-based email clients / webmail include: **Gmail, Yahoo!, Lycos mail, and Hotmail**.

In order to send and receive an Email, both the sender and the recipient need the following; -

1. Internet connection
2. Email address
3. Email client: This is an application program or software that users can use to access and manage email accounts. Today, this email client maybe web based or not

Email Address

An email address is a string of characters which identifies the location on the internet or other network to which an email can be sent.

This string is considered as a point of contact for a particular individual, organization, etc. Like social networking sites use user names, email software uses email address.

Parts of an Email address

username@domain name. top level domain

Consider the email address: **mubirurodney68@gmail.com**

1. **Username:** This is a nickname/ handle of an individual or organization. In our example; the username is Mubirurodney.
2. **Domain name:** This refers to the name of the computer on which the email account resides. It is the mail server. In our example; the domain name is Gmail, which stands for Google mail.
3. **Top level domain:** This specifies the purpose of the host. In our example; the top-level domain is .com which means that the host is a commercial enterprise trying to make money.

Top level domain name

purpose of hosting

.com	:	Commercial or business enterprise
.org	:	Non- commercial organization
.edu	:	Educational institution
.ac	:	Academic institution
.mil	:	Military
.net	:	Network company or ISP
.gov	:	Government

Some email addresses have a [Dot] followed by two letters. These letters usually represent the country and are called country codes

Country-Code Top-Level Domains (CCTLDS)

cTLD	Country	Example
.us	United States	www.example.us
.uk	United Kingdom	www.bbc.co.uk
.ca	Canada	www.example.ca
.au	Australia	www.example.au
.in	India	www.example.in
.ke	Kenya	www.example.ke
.za	South Africa	www.example.za
.ug	Uganda	www.example.ac.ug

Common Second-Level Domains (SLDs) Used in Some Countries

Some countries use **structured second-level domains** before the main top-level domain:

Second-Level Domain (SLD)	TLD	Purpose	Example
.co	.uk	Commercial entities in the UK	www.bbc.co.uk
.ac	.uk	Academic institutions in the UK	www.cam.ac.uk
.gov	.uk	Government institutions in the UK	www.gov.uk
.edu	.au	Educational institutions in Australia	www.unimelb.edu.au
.co	.ke	Commercial organizations in Kenya	www.example.co.ke
.ac	.ug	Academic Institution in Uganda	www.must.ac.ug

Parts of an Email message

An email message has got three parts namely

1. **Message header:** The message headers contain information concerning the sender and recipients. The exact content of mail headers can vary depending on the email system that generated the message. Generally, headers contain the following information

Features of the headers

From: This contains the Email address of the person sending the message.

To: This contains the Email address of the person intended to receive the message.

Date: This contains the date and time on which the Email has been sent.

Subject: This contains the phrase or statement summarizing the main theme or topic of the message. For example; APPLICATION, JOB PROPOSAL, among others.

CC (Carbon Copy): This feature allows the user to send the same message to different recipients with each of the recipients noticing.

BCC (Blind Carbon Copy): This feature allows the user to send the same message to different recipients but without each of the recipients noticing.

2. **Body:**

The body of a message contains text that is the actual content.

The message body also may include signatures or automatically generated text that is inserted by the sender's email system.

3. **Attachments**

Attachments are optional and include any separate files that may be part of the message.

E-mail Vocabulary

- **Sign up** (Create Account): Registering your email account details
- **Login** or Sign in: Opening your email account using your account details (username and password)
- **Logout** or Sign Out: Closing your email account
- **Inbox**: Folder for incoming messages
- **Sent** or Outbox: Folder for outgoing messages
- **Trash**: Folder for deleted messages
- **Compose** or New: Write a new message
- **Drafts**: A folder for all typed but not sent messages
- **Outbox**: A folder for all messages that are sent but didn't go

Advantages of using email as a means of communication

- Easy to use. Email applications have user friendly tools that help during composing messages.
- Email supports sending of attachments like documents, zipped files, etc.
- It is very fast in terms of speed: The e-mail is delivered instantly, anywhere across the globe.
- Easy to prioritize: Since the mails have subject lines, it is easy to prioritize them and ignore unwanted mails.
- Email messages can be sent to many recipients at the same time
- Emails can also carry hyperlinks that lead to other webpages with just a click
- One can subscribe to news and other online services through email
- Email software has management features that help users to organize their messages in folders like inbox, sent, draft, etc.
- Easier for reference: When one needs to reply to a mail, there is a provision in the mailing system to attach the previous mails as references. This refreshes the recipient's knowledge on what he is reading.
- Environment friendly: Compared to postal mails which use paper and fuel to transport letters. Electronic mail saves a lot of trees from being axed. It also saves fuel needed in transportation.
- Email software has address book features that may be sorted in alphabetical order.
- Email software has a good degree of security features such as username and password before sign in
- Email applications have an inbuilt English dictionary which safeguards the sender from incorrect spelling and grammar.
- Email is a relatively cheap means of communication since there are no printing or postage expenses involved.
- 24/7 anytime access: At any time of the day or night, one can communicate with friends, relatives, professors and business associates.
- Messages remain permanent for future access from anywhere.
- Use of graphics such as colorful greeting cards and interesting pictures can be sent through emails.
- Advertising tool: many individuals and companies are using e-mails to advertise their products, services, etc.

Limitations of using Email as means of communication.

Emails can spread Viruses: The recipient needs to scan the mails, as viruses are transmitted through them and have the potential to harm computer systems.

Spam and Junk: E-mails when used to send unsolicited messages and unwanted advertisements create a nuisance called Spam. Checking and deleting these unwanted mails can unnecessarily consume a lot of time, and it has become necessary to block or filter the unwanted emails by means of spam filters.

E-mail spoofing is another common practice. Spoofing involves disguising as a different sender by altering the e-mail headers or the addresses from which the mail is sent.

Hacking and email interception: The act of unauthorized attempts to bypass the security mechanisms of an information system or network is termed as hacking. After the email is sent and before it is received by the desired recipient, it "bounces" between servers located in different parts of the world. Hence, the e-mail can be intercepted by a professional hacker.

Misinterpretation: One has to be careful while posting any kind of content through an email. If typed in a hurry, the matter could be misinterpreted. Since the content posted via e-mails is considered informal, there is a chance of business documents going unnoticed. Thus, vital communications and especially those requiring signatures are not managed through emails

Crowded inbox: Over a period of time, the email inbox may get crowded with mails. It becomes difficult for the user to manage such a huge chunk of mail.

Need to check the inbox regularly: In order to be updated, one has to check his email account regularly, which may be expensive in the long run.

Email cannot be used without computers especially in remote areas without electricity.

In case **one forgets his/her** password, signing in is not possible and this can lead to loss of information.

Email may violate privacy in case someone else gets to know your user password since the other may check your mails.

Using cloud services to share information

Cloud Computing

Cloud computing is Internet-based computing, whereby shared resources e.g hardware, software and information are provided to other devices on-demand.

In simple terms, Cloud computing is using the internet to access someone else's software running on someone else's hardware in someone else's data center.

Cloud computing operates on a similar principle as web-based email clients, allowing users to access all of the features and files of the system without having to keep the bulk of that system on their own computers. In fact, most people already use a variety of cloud computing services without even realizing it such as Gmail, Google Drive, Google Docs, etc.

Advantages of Cloud Computing

- **Lower computer costs:**
You do not need a high-powered and high-priced computer to run cloud computing web-based applications.
Since applications run in the cloud, not on the desktop PC, your desktop PC does not need the processing power or hard disk space demanded by traditional desktop software.
- **Improved performance:** With few large programs hogging your computer's memory, you will see better performance from your PC.

- **Reduced software costs:** Instead of purchasing expensive software applications, you can get most of what you need cheaply, e.g most cloud computing applications today, such as the Google Docs suite, better than paying for similar commercial software.
- **Better Security:** By using encryption, information on the cloud is less accessible by hackers or anyone not authorized to view the data. As an added security measure, with most cloud-based services, different security settings can be set based on the user.
- **Instant software updates:**
Another advantage to cloud computing is that you are no longer faced with choosing between obsolete software and high upgrade costs.
When the application is web-based, updates happen automatically.
- **Improved document format compatibility**
You do not have to worry about the documents you create on your machine being compatible with other users' applications.
- **Unlimited storage capacity:** Cloud computing offers virtually limitless storage on servers in powerful datacenters.
- **Increased data reliability/ safety:** Unlike desktop computing, in which if a hard disk crashes and destroys all your valuable data, a computer crashing in the cloud should not affect the storage of your data.
- **Universal document access:** That is not a problem with cloud computing, because you can access it whenever you have a computer and an Internet connection.
- **Easier group collaboration:** multiple users can collaborate easily on documents and projects
- **Device independence:** Even to a portable device, your applications and documents are still available

Disadvantages of Cloud Computing

- **Requires a constant Internet connection:** Cloud computing is impossible if you cannot connect to the Internet. A dead Internet connection means no work.
- **Does not work well with low-speed connections:** Web-based applications require a lot of bandwidth to download, as do large documents
- **Features might be limited:** Many web-based applications simply are not as full-featured as their desktop-based applications. For example, you can do a lot more with Microsoft PowerPoint than with Google Presentation's web-based offering.
- **Can be slow-**Even with a fast connection, web-based applications can sometimes be slower than accessing a similar software program on your desktop PC. Everything about the program, from the interface to the current document, has to be sent back and forth from your computer to the computers in the cloud.
- **Stored data might not be secure:** With cloud computing, all your data is stored on the cloud. Any unauthorized users gaining access to your password may access confidential data.
- **Migration issues:** Each cloud system uses different protocols and different APIs, so your normal applications will have to be adapted to execute on these platforms.

Cloud computing Services

- | | |
|--|-----------------------------|
| ✓ iCloud (for Apple) | ✓ DropBox |
| ✓ OneDrive (Formerly called SkyDrive
– for Microsoft) | ✓ HighTail |
| ✓ IBM Smart Cloud (for IBM) | ✓ SugarSync |
| | ✓ Google Drive (for Google) |

Using Networks to Share Information

A computer network is defined as a collection of computers linked together using transmission media for the purpose of communication and resource sharing.

Some of the shared resources include internet connectivity, printers, fax machines, modems, storage devices, networked software programs etc.

Basic requirements for setting up a computer network

- ✓ A set of computers installed with an operating system such as Windows or Linux
- ✓ Each computer should have a LAN card (Network Interface Card- NIC).
- ✓ Ethernet cables, crimped with RJ-45 connectors
- ✓ Network Switch or Hub depending on the nature of the network desired
- ✓ Router/Modem (for internet connections)

Resources that can be shared on a school network

- ✓ File and folders or documents such as invoices and budgets
- ✓ Storage space on a hard disk
- ✓ Programs such as Encarta or Microsoft Applications
- ✓ Optical drives such as DVD/CD ROM drives
- ✓ Multimedia information such as videos, audio files, photographs and maps
- ✓ Email
- ✓ Scanner
- ✓ Printers

Application that supports file sharing

- | | |
|-------------|-------------|
| ✓ WhatsApp | ✓ Instagram |
| ✓ Facebook | ✓ Twitter/X |
| ✓ Snap Chat | |

Advantages of computer networks in schools

- ✓ Cost saving because of sharing network resources. For example, one printer can be connected to a network and used by 10 or more people.
- ✓ It improves communication because staff can easily share information conveniently
- ✓ File sharing: Users on the same network can easily share documents
- ✓ Resource sharing: Users on the same network can easily share network-connected peripheral devices like printers and scanners.
- ✓ Sharing a single internet connection. Users on the same network can share internet access
- ✓ Increased storage space. Users on the same network can easily access files and multimedia, such as images and music, stored remotely on other machines or network-attached storage devices.

Disadvantages of computer networks in schools

- ✓ Setting up network infrastructure is expensive in the short run because of the need to buy hardware and cables
- ✓ If a hacker gets unauthorized access to the network, then there are higher chances that all computers on the network would be affected.
- ✓ If one computer on a network gets infected by a worm or virus, then there are higher chances that all computers on the network would be affected
- ✓ If the network server fails, then all computers on the network will not be able to access information on the network.
- ✓ Requires Administrative Time. Proper maintenance of a network requires considerable time and expertise

- ✓ **Moral and cultural effects:** large networks like the Internet have chat rooms and messaging services that may enable underage children to meet peers and adults on the net some of whom may have bad intentions
- ✓ **Spread of terrorism and drug trafficking:** The Internet provides a rich recruitment ground for all types of illegal activities such as terrorism and drug trafficking. The easy flow of information from one place to another keeps even those who are on the wrong side of the law communicating easily.

Types of computer networks

- ✓ **Local Area Network:** Covers a small geographical area such as a home, office, building or room
- ✓ **Personal Area Network:** A computer network that connects devices within a limited range, typically 10m.
- ✓ **Campus Area Network:** A computer network that connects devices within a college or university.
- ✓ **Metropolitan Area Network:** Covers a large area such as a city or town
- ✓ **Wide Area Network:** Covers a vast geographical area such as cellular network.
- ✓ **Virtual Private Network:** is a secure, encrypted connection between two or more networks over the internet
- ✓ **Intranet:** A private network within an organization using technologies to share information and resources.
- ✓ **Extranet:** A private network that allows external partners, customers or suppliers to access specific resources or information within an organization.

Sharing Information using Bluetooth

Bluetooth is a short-range wireless based information transmission system which works on the basis of microchips embedded in the digital devices like mobile phones, speakers and laptops.

Advantages of Bluetooth Technology

- Bluetooth technology is free to use
- It provides a quick connection that does not require wires
- Its range in distance is better than infrared communication
- It has no issue of interoperability among different Bluetooth devices
- It can be used for both voice and data transfer
- Most Bluetooth devices are affordable

Disadvantages of Bluetooth Technology

- It works for only short-range data or voice transfer
- It provides low bandwidth compared to Wi-Fi
- It can easily be hacked because it operates on radio frequency

Safety Practices while Using Internet

The internet is a good place with a variety of information required by all people. However, the internet can be dangerous if safety precautions are not followed.

Below are some of the safety practices to follow while using Internet.

Avoid posting personal information on websites, blogs, social networking sites and other places on the internet. The information may be used as a starting point by malicious people to harm you

Always use very strong passwords that are hard to crack

Internet Safety (Online Safety)

Internet Safety is the action of staying safe when you are online. It involves the user's awareness of personal safety and security risks involving private information

Safety Practices while using Internet

- Ensure that the internet connection is secure. Do not connect to public Wi-Fi that is not password protected
- Mind about what you download, not everything is good for you and your computer
- Use strong passwords containing numbers, letters, signs, symbols, lowercase and uppercase. The password should be at least 10 characters long.
- Be careful with what you post. Avoid posting personal information on websites or social media which may be used by malicious people
- Keep privacy settings on and Safe browsing on when using web browsers and mobile operating systems on mobile devices. This will protect you from hackers
- Make online payments and purchases from only trusted websites. Hackers can steal credit card or bank account details if you transact using insecure websites.
- Install anti-virus programs on your computer and have it regularly updated. Scan all devices you use.
- Do not open emails if you are not sure of the senders. Some emails are from people with malicious intent and they may carry malware which will harm your computer once opened.

Areas of Internet that need safety

- Online banking
- Email communication
- E-commerce
- Online learning
- Social media like WhatsApp, Facebook etc.

Possible rules to follow while using social media like WhatsApp on Smartphones at School

- All phones must be off during class hours unless the teachers advise otherwise
- Nobody should use social media for more than a lesson
- If you have social media apps on your Smartphone, be sure its password protected
- Be selective with friend requests

Netiquette

Netiquette is a social code of internet use on computers, mobile phones and handheld devices. Netiquette is a set of rules on networks or websites.

General Netiquette Guidelines

Rule	Explanation
Be Respectful	Treat others with kindness, just as you would in person. Avoid rude or offensive language.
Think Before You Post	Consider the impact of your words or images before sharing them online.
Use Proper Grammar & Spelling	Avoid excessive use of slang, caps lock (SHOUTING), or text abbreviations in formal communication.

Avoid Spamming	Do not send excessive messages, advertisements, or irrelevant content.
Respect Privacy	Do not share personal or sensitive information about yourself or others.
Be Cautious with Humor & Sarcasm	Tone is hard to convey in text, so ensure jokes are not offensive or misleading.

Email & Messaging Netiquette

Rule	Explanation
Use a Clear Subject Line	Helps recipients understand the purpose of the email.
Greet and Sign Off Politely	Start with “Dear [Name]” and end with “Best regards” or “Thank you.”
Reply Promptly	Respond to emails/messages in a timely manner.
Avoid Overusing CC/BCC	Only include relevant recipients.
Keep Messages Short & Clear	Long messages can be overwhelming; be direct and professional.

Social Media Netiquette

Rule	Explanation
Think Before Commenting	Avoid negative, offensive, or misleading comments.
Give Proper Credit	If sharing content, credit the original creator.
Respect Opinions	Discussions should be civil, even if you disagree.
Avoid Excessive Tagging	Only tag people if necessary.
Do Not Overshare Personal Details	Protect your privacy by limiting personal information.

Online Classroom & Video Call Netiquette

Rule	Explanation
Mute When Not Speaking	Avoid background noise during virtual meetings.
Dress Appropriately	Wear suitable attire for online classes or meetings.
Be Punctual	Join meetings on time and avoid distractions.
Use Professional Language	Speak clearly and avoid slang.
Participate Actively	Engage in discussions but avoid interrupting others.

Cybersecurity & Ethical Netiquette

Rule	Explanation
Do Not Share Fake News	Verify facts before spreading information.

Avoid Cyberbullying	Be kind and report any harmful behavior online.
Use Strong Passwords	Protect your accounts from hacking.
Do Not Plagiarize	Give proper credit when using someone else's work.
Respect Copyright Laws	Do not illegally download or share copyrighted material.

Possible Risks Associated with Internet Use

- ✓ Financial loss: Malicious people can use malware to steal information on credit cards and debit cards, and bank account details
- ✓ High risk of physical harm, harassment and sexual abuse by persons
- ✓ Identity theft. The internet has many websites that ask for personal information in order to assume the identity of the victim.
- ✓ Exposure to illegal and harmful content, such as pornography, obscene games and contact with unknown people.
- ✓ Exposure to drug related information for example about marijuana, tobacco and alcohol
- ✓ Possible damage to the image of the company, especially when an employee posts the company's confidential information
- ✓ Loss of productivity, when employees waste time chatting on social media or playing games.
- ✓ Violation of copyright. This may happen when people download music, movies and software, and use them for commercial purposes without paying for them.
- ✓ Computer malware such as viruses, Trojan horses, spyware and adware may lead to computer malfunctions and total loss of essential information
- ✓ Exposure to racist and discriminatory language, texts and images which all have a negative effect on children.

Laws on Information Access and Sharing

Many countries, Uganda inclusive have laws to control people who misuse computers and the internet. In Uganda, we have the Computer Misuse Act.

HEALTH AND SAFETY

Health and safety in ICT refers to practices and precautions taken to protect users from physical and mental harm while using information and communication technologies. Since ICT use often involves prolonged screen time, sitting posture, and repetitive actions, it's important to understand how to prevent injuries and maintain a safe working environment.

People spend several hours in front of a computer without thinking about the impact on their bodies.

They physically strain their bodies without realizing it by extending their wrist, bending, sitting without foot support and straining to look at poorly placed monitors.

This practice can lead to cumulative trauma disorders or repetitive strain injuries, which create a life-long impact on health.

Symptoms may include pain, muscle fatigue, loss of sensation, tingling and reduced performance.

Humans are supposed to use computers with regular breaks. Another way of preventing the above is by using computer ergonomics.

Health problems associated with prolonged use of computers/ICT tools (devices)

Deep vein thrombosis

This is a medical condition which involves a clot forming in the leg vein. It is life threatening because the small bits of the clot can break off, travel through the blood system and lodge in the brain, resulting in a stroke.

Cause of deep vein thrombosis

DVT in the context of computing can develop due to prolonged sitting which leads to reduced blood flow.

Sitting for a long time without legs support

Symptoms

- Swelling in the affected leg
- Pain in the affected leg
- Tenderness
- Redness in the leg

Ways to Avoid Deep Vein Thrombosis

- Regularly stand up and move around
- Perform leg exercises while sitting
- Use an adjustable desk to alternate between sitting and standing
- Stay hydrated.

Repetitive Strain Injury

Overusing computers can bring about a painful swelling on the wrist, fingers and others parts of the body due to repeated small movements, such as repetitive mouse click.

Causes of Repetitive Strain Injury (RSI)

- Typing for a long period of time
- Using too much force while typing
- Poor ergonomics or a non-ergonomic workstation setup.

Symptoms

- Swelling of the arms and hands
- Aching of the arms and hands

- Stiffness in the affected areas
- Throbbing in the affected areas
- tingling or numbness in the affected areas
- weakness, or cramps in the affected areas (often the hands, wrists, forearms, shoulders, or neck).

Ways to avoid RSI

- Take regular breaks from typing to rest the hands using the 20-20-20 rule (every 20 minutes, take a 20-second break and look at something 20 feet away),
- Use wrist rests
- Ensure an ergonomic workstation setup
- Practice good sitting posture

Back and neck pain

Prolonged use of computers can bring about back and neck pains especially due to poor sitting posture and incorrect positioning of the computer screen.

Cause of Back and Neck pain

- Poor sitting posture
- Poor positioning of computer screen (monitors that are not at eye level causing one to lean forward or slouch)
- Non-ergonomic chairs
- Working in a cramped workspace. A cramped workspace refers to an area where someone works that is uncomfortably small or restricted, offering limited space to move around or arrange necessary equipment and resources.

Symptoms

- Back pain while walking, standing and bending
- Pain when moving the neck
- Pain in the lower back, upper back, shoulders, or neck.
- Discomfort
- Stiffness

Ways to avoid Back and neck pain

- Use adjustable chairs for better heights depending on the computer desk
- Use an ergonomic chair that supports the spinal curves
- keep the monitor at eye level and at an arm's length away
- Ensure that feet are flat on the ground or on a footrest sit upright
- Do not bend your head
- Take frequent breaks to move and stretch.

Eye strain and headache

Prolonged use of computers can cause eye strain and headache. This can result in blurry eye vision.

Causes of Eye Strain and headache

- Looking at a computer screen for a long period of time
- Working in a room with dim light
- Glare on the screen
- Using devices with small text

Symptoms of eye strain and headache

- | | |
|-----------------|----------------|
| - Itching eyes | - Headache |
| - Blurry vision | - Irritability |

Ways to avoid eye strain and headache

- Follow the 20-20-20 rule
- Adjust the computer display settings (brightness, text size, contrast)
- Ensure proper lighting to reduce glare
- Consider using anti-glare screens or glasses.

Stress: Causes

- Stress related to computing can stem from prolonged periods of concentrated work without breaks,
- Pressure to meet deadlines,
- Multitasking,
- Dealing with complex technical issues.

Symptoms of Stress

- Symptoms include feelings of overwhelm,
- Irritability
- Anxiety
- Difficulty concentrating,
- Insomnia
- Low productivity
- Illness and fatigue.

Ways to avoid stress

- Manage stress by taking breaks
- Practicing time management
- Setting realistic goals
- Maintaining a clean and organized work environment
- Using relaxation techniques such as deep breathing or meditation.
- Switch off some ICT tools while at home to avoid distractions

Suffocation

- **Suffocation:** ICT Equipment generates a lot of heat as user do users
- **Causes:** Accidental Entrapment, Equipment Malfunction, too much heat generated by ICT tools and users and Poor Ventilation
- **Symptoms:** Loss of breath, Headaches, Confusion or Dizziness and Loss of Consciousness

Ways to avoid suffocation

- Ensure proper ventilation of your work station
- Make sure all the ICT equipment are in a good working condition

What is the 20-20-20 rule?

- Every 20 minutes, take a break from your screen.
- Look at something 20 feet away (approximately 6 meters).
- Do this for at least 20 seconds.

This rule is intended to give your eyes a chance to rest and refocus, which can help prevent the fatigue and discomfort associated with staring at digital displays for too long. It helps to relax the eye muscles and can reduce the risk of headaches, blurred vision, and other symptoms of digital eye strain.

Computer ergonomics

Computer ergonomics, also known as office ergonomics, refers to the design and arrangement of the workplace to fit the worker, aiming to enhance efficiency while reducing discomfort and

the risk of injury. Proper computer ergonomics encompasses the setup of computer equipment and the environment in which people work.

Here are some key components to consider:

Chair

- Choose a chair that supports your spinal curves.
- Adjust the height so your feet are flat on the floor or on a footrest, and your thighs are parallel to the floor.
- Ensure that your forearms are parallel to the ground when on the keyboard.

Desk

- Ensure that the desk height is such that your elbows are at a 90-degree angle when typing.
- There should be enough space under the desk to comfortably stretch your legs.
- Consider a sit-stand desk that allows you to alternate between sitting and standing.

Monitor

- Position the monitor directly in front of you, about an arm's length away.
- The top of the screen should be at or slightly below eye level.
- Use an adjustable stand if needed to achieve the ideal height and angle.

Keyboard and Mouse

- Place the keyboard directly in front of you when typing. Leave a gap of about 4 to 6 inches at the front of the desk to rest your wrists when not typing.
- Use a keyboard tray to achieve proper positioning if needed.
- The mouse should be placed close to the keyboard so it can be reached with a slight bend in the elbow.
- Consider using ergonomic keyboards and mouse devices that provide better comfort.

Lighting

- Ensure adequate lighting to avoid eye strain.
- Position the computer screen to avoid glare from overhead lighting or windows.
- Use task lighting if needed to have adequate light on documents or other items on your desk.

Breaks and Movement

- Take short breaks regularly to stand, stretch, and walk around.
- Practice the 20-20-20 rule to give your eyes a break from the screen.

Work Habits

- Organize your workspace so that frequently used items are within easy reach.
- Avoid prolonged static postures by regularly changing your position.
- Practice good posture: Keep your head and neck upright and in-line with your torso.

Poor sitting posture

- Sitting with the upper back curved
- Sitting without resting the feet on the floor
- Sitting while crossing knees or ankles
- Sitting while straining the neck to look up or down

When using a desktop computer, the user should ensure that;

- the top of the monitor is at level of their eyes
- the chair they sit on completely supports their thighs
- the legs are bent at 90-100-degree angle
- their feet are flat on the floor
- their shoulders are relaxed
- the chair has a backrest that supports their lower back

WORD PROCESSING II

Advanced Features of Word Processors

Modern word processing software, such as Microsoft Word, LibreOffice Writer, and Google Docs, provide advanced features that enhance the appearance, structure, and functionality of documents.

Text Wrapping

Text wrapping controls how text flows around objects like images, charts, or shapes.

Types

- | | |
|---------------------|------------------|
| - In line with text | - Through |
| - Square | - Top and bottom |
| - Tight | |

Text wrapping improves the visual layout when combining text with images or diagrams.

Cropping Images

Cropping an image refers to the process of removing or trimming away the unwanted outer parts of a picture. This is done to focus attention on a specific area of the image or to remove unnecessary background elements. When you crop an image, you reduce its visible area without changing its original resolution. The cropped part is hidden from view but can be restored later if needed.

Compressing images

Compressing images means reducing the file size of the pictures inserted into a document. This is especially useful when working with large documents that contain many images or when the document needs to be shared via email or uploaded online. Compression works by lowering the resolution or quality of the images to an acceptable level while keeping them visually clear enough for their intended use.

Multiple Columns

Splits text into two or more vertical sections within a page.
Commonly used in newsletters, magazines, and brochures.

Footnotes and Endnotes

Notes placed at the bottom of a page (footnote) or at the end of a document (endnote) to provide additional information or references.

Useful in academic writing, reports, and research papers.

Watermark

A faint text or image behind the document content.

To indicate confidentiality (e.g., "Draft", "Confidential") or branding purposes.

Drop Cap

A large capital letter at the beginning of a paragraph that drops down two or more lines.

Adds a decorative touch to the beginning of a section or chapter.

Table of Contents (TOC)

An automatically generated list of headings and subheadings in a document.

Use: Provides a structured overview and makes navigation easier in long documents.

List of Figures and Tables

Similar to the TOC, these lists automatically display the titles and page numbers of all figures or tables in a document.

Enhances organization and accessibility in academic and technical documents.

Captions

Labels added below or above images, tables, or figures.

Provides descriptions or titles, often used in conjunction with lists of figures/tables.

Password Protection

A security feature that restricts access to a document by requiring a password to open or edit it.

Protects sensitive or confidential information from unauthorized access.

Mail Merge

A feature that allows you to generate multiple personalized documents (like letters, certificates, or labels) from a single template using data from a source file (e.g., Excel).

Saves time when sending the same document to many people with slight variations (like names or addresses).

Section and Page Breaks

Page Break

A command that forces the text following it to begin on a new page.

Useful when separating content logically, like starting a new chapter or topic on a fresh page.

Section Break

A break that divides a document into sections, allowing each section to have different formatting (e.g., margins, orientation, header/footer layout).

Types

- Next Page – Starts the new section on the next page.
- Continuous – Starts the new section on the same page.
- Even/Odd Page – Begins the section on the next even or odd-numbered page.

Ideal for creating documents with varied layouts, such as mixing portrait and landscape orientations or using different page numbering styles.

Page Numbers

Numbers automatically assigned to pages in a document, usually shown in headers or footers.

Formats

- Arabic (1, 2, 3)
- Roman (i, ii, iii)
- Letters (A, B, C)

Helps readers navigate through the document and is essential for academic, business, or formal documents.

With section breaks, you can restart page numbering or use different formats in each section.

Equations

An equation is a mathematical expression created and formatted using special tools designed to represent complex symbols, formulas, and structures clearly and professionally. Word includes a built-in Equation Editor that allows users to insert standard or custom mathematical equations into their documents.

Charts

A chart in MS Word is a visual representation of data, commonly used to make information clearer and easier to understand. Charts are useful when comparing values, showing trends over time, or displaying relationships in data.

SPREADSHEET II

Filtering data

Filtering is the process of displaying only the rows in a spreadsheet that meet certain conditions, while hiding the others. This helps you focus on specific information in large datasets without deleting any data.

How to Filter Data in Microsoft Excel

- Open your spreadsheet.
- Click on any cell inside your data table.
- Go to the Home or Data tab on the Ribbon.
- Click Filter (found in the Sort & Filter group).
- Small dropdown arrows (▼) will appear at the top of each column.
- Click a dropdown arrow to:
 - Select or deselect specific values to show.
 - Use text, number, or date filters (e.g. “Greater than...”, “Contains...”).

Filter Types

Type	Use Example
Text Filter	Filter names that start with “J” or contain “son”
Number Filter	Show only values > 50 or between 10 and 100
Date Filter	Filter data from a specific month or year
Color Filter	Filter cells with a certain background or font color

Cell References

A cell reference is a cell or range of cells on a worksheet and can be used in a formula

A cell reference in a spreadsheet (like Excel or Google Sheets) refers to the address or location of a cell within the worksheet. It tells the spreadsheet where to find the data you want to use in a formula or function.

Example: A1 refers to the cell in Column A, Row 1.

Types of cell references

Relative Cell Referencing

A relative cell reference adjusts automatically when copied or filled to another location. It is the default reference type in spreadsheets.

Example:

- If cell D1 has the formula =A1+B1
- When copied to D2, it becomes =A2+B2

Absolute Cell Referencing

An absolute reference does not change when the formula is moved or copied. It always points to the same specific cell.

Example:

In cell C2, if the formula is =A2*\$B\$1 and you copy it to C3, it becomes =A3*\$B\$1 (Only A2 changed; \$B\$1 remains fixed)

Mixed Cell Referencing

A mixed reference fixes either the row or the column, but not both.

Example:

- Formula in cell B2: =A\$1
- When copied to C3, it becomes =B\$1

Page Setup

Page Setup refers to adjusting the layout and formatting of a document's page. It controls how the content fits on a page when printed or viewed.

Example: Setting Up a Worksheet for Printing

- Set margins: Go to Page Layout > Margins > Normal (or customize).
- Change orientation: Page Layout > Orientation > Landscape (if your table is wide).
- Set print area: Select the cells you want to print, then Page Layout > Print Area > Set Print Area.
- Repeat row headings: Page Layout > Print Titles > specify rows to repeat at top.
- Fit sheet to one page: Page Layout > Scale to Fit > set Width and Height to 1 page.

Key Page Setup Features

Option	What It Does
--------	--------------

Margins	Adjusts space around the printed page (top, bottom, left, right).
Orientation	Choose Portrait (vertical) or Landscape (horizontal) layout.
Size	Select the paper size (e.g., Letter, A4).
Print Area	Defines the specific cells or range to print.
Print Titles	Choose rows or columns to repeat on every printed page (useful for headings).
Scaling	Adjust the size of the worksheet to fit on one page or multiple pages.

Excel Error Messages

Error	The meaning	Example or cause	Fixing the error
#####	The column width is not wide enough to display the data or a value	A calculation has resulted in a very long result	Widen the column (this is not an error)
#VALUE!	- You have omitted a function name or values in a formula - You have typed formula leaving out brackets - Used a wrong operator or data type - Referencing to text by numeric operations	You have specified a mathematical operation with one or more cells that contain text. A function is not specified for example using: +(b2:c2)	Correct the data entry by checking: - Function name - The operands and arguments - And make sure that references are valid
#DIV/0!	The formula is trying to divide by zero. For example a formula; = D2/0 gives division error	Division operation in your formula refers to a cell that contains zero or is blank	Replace the zero value in the cell with a non-zero number
#NAME?	- Function name in the formula is not recognized or - You have entered text in a formula without enclosing it in quotation marks	You have referenced to a range incorrectly For example, writing the formula as =(SAM A2:A4) instead of =SUM(A2:A4)	- Correct spelling mistakes of the function name or - Enclose text in quotation marks or - Ensure the range name exists
#NUM!	A formula has invalid numeric data for the type of operation, for example, using the formula = sqrt(-16)	Either the argument is invalid or the result is too large or small to display	Type arguments correctly
#REF!	The reference used is invalid. For example, cell, sheet, row or column has been deleted	- You have deleted a cell referred to in a formula - You have pasted cells over the ones rereferred to in the formula	- Correct all cell references - Click undo to reverse the wrong action done

#N/A	Values cannot be found in referenced data. For example, referencing "Banana" in the lookup table which does not exist, the VLOOKUP returns a N/A error	This is not an error value but special value entered to indicate you do not have a necessary value yet	- Type in the values - Correct the values with correct arguments
#NULL	A special character was used instead of a comma (,) or colon (:) between two cell references or ranges. E.g., D2 D7 instead of D2:D7 or D2,D7. You specified an intersection between two ranges that do not intersect such as B1:D3 and E4:G6 (B1:D3 E4:G6) instead of (B1:D3,E4:G6)	If you insert a space instead of comma between ranges used in function arguments. Note: Space indicates an intersection. A comma is a union operator	Check for typing errors and referencing errors

Advanced Excel Functions

NOTE: Each formula or function **must** start with an **equal sign (=)**

SUM

Syntax: SUM (number1, [number2], ...)

Example: SUM (A1:A10) - Adds all numbers in the range A1 to A10.

AVERAGE

Syntax: AVERAGE (number1, [number2], ...)

Example: AVERAGE (B1:B10) - Calculates the average of the numbers in the range B1 to B10.

MIN

Syntax: MIN (number1, [number2], ...)

Example: MIN (C1:C10) - Returns the smallest number in the range C1 to C10.

MAX

Syntax: MAX (number1, [number2], ...)

Example: MAX (D1:D10) - Returns the largest number in the range D1 to D10.

COUNT

Syntax: COUNT (value1, [value2], ...)

Example: COUNT (E1:E10) - Counts the number of cells that contain numbers in the range E1 to E10

COUNTA

Syntax: =COUNTA (value1, [value2], ...)

Example: =COUNTA (A1:A10) - Counts all the cells (whether blank or containing data) in the range **A1:A10**

IF

Syntax: IF (logical_test, value_if_true, value_if_false)

Example: =IF (A1 > 10, "Yes", "No")

Explanation: Checks if the value in cell A1 is greater than 10. If true, returns "Yes"; otherwise, returns "No".

VLOOKUP

Syntax: VLOOKUP (lookup_value, table_array, col_index_num, [range_lookup])

Example: =VLOOKUP ("Apple", A1:B10, 2, FALSE)

Explanation: Looks for "Apple" in the first column of the range A1 and returns the value in the second column of the found row. The FALSE argument ensures an exact match.

HLOOKUP

Syntax: HLOOKUP (lookup_value, table_array, col_index_num, [range_lookup])

Example: =HLOOKUP ("Price", A1:D4, 3, TRUE)

Explanation: Looks for "Price" in the first row of the range A1 and returns the value in the third row of the found column. The TRUE argument allows for an approximate match.

INDEX

Syntax: INDEX (array, row_num, [column_num])

Example: =INDEX (A1:C3, 2, 3)

Explanation: Returns the value in the second row and third column of the range A1

MATCH

Syntax: MATCH (lookup_value, lookup_array, [match_type])

Example: =MATCH (20, B1:B10, 0)

Explanation: Searches for the value 20 in the range B1 and returns the relative position of the value. The 0-match type requires an exact match.

COUNTIF

Syntax: COUNTIF (range, criteria)

Example =COUNTIF (A1:A10, ">5")

Explanation: Counts the number of cells in the range A1 that are greater than 5.

SUMIF

Syntax: SUMIF (range, criteria, [sum_range])

Example: =SUMIF (B1:B10, ">=20", C1:C10)

Explanation: Sums the values in the range C1 where the corresponding cells in the range B1 are greater than or equal to 20. If sum_range is not provided, range is summed instead.

Other Excel Functions

SQRT: This returns the square root of any specified number

Syntax: =SQRT(number) e.g., =SQRT(100).
The answer is 10

COUNTA: This counts the number of cells which are not empty in the specified range.

Nested IF

Syntax: IF (logical_test1, value_if_true1, IF (logical_test2, value_if_true2, value_if_false2))

Example: =IF (A1 > 90, "A", IF (A1 > 80, "B", IF (A1 > 70, "C", "D")))

Explanation: This nested IF function assigns grades based on the value in cell A1:

If A1 > 90, the result is "A".

If A1 is not greater than 90 but is greater than 80, the result is "B".

If A1 is not greater than 80 but is greater than 70, the result is "C"

If A1 is not greater than 70, the result is "D".

LARGE

Syntax: LARGE (array, k)

Example: =LARGE (A1:A10,1)

Explanation: Returns the first largest number in the range A1:A10

SMALL

Syntax: SMALL (array, k)

Example: =SMALL (B1:B13, 2)

Explanation: Returns the second smallest number in the range B1 to B13

Syntax: =COUNT(range) e.g.,
=COUNTA(B3:C12)

COUNTBLANK: This counts empty cells in a specified range. Cells with zeros are not counted.

Syntax: =COUNTBLANK(range) e.g.,
=COUNTBLANK(B3:C12)

TODAY: This returns the current date (internally as a serial number) i.e days that have elapsed since 1st Jan 1900. This takes no arguments.

Syntax: =Today(); the result may be 5/17/2025.

NOW: This returns the current time. It requires no arguments

Syntax: =Now() the answer May 21, 2025 12:00

MODE: This displays the mode or most frequent number in a range of cells.

Syntax: =MODE(range) e.g., =Mode(A1:B9).

MEDIAN: This displays the median of the numbers in all cells specified or range

Syntax: =MEDIAN(range)

Freeze or lock rows and Columns

Freezing rows or columns keeps specific headers or labels visible while scrolling, making large spreadsheets easier to read.

How to Freeze Panes in Excel

Freeze Top Row

Use when your spreadsheet has a header row (e.g., titles like Name, Age, Score).

- Go to the View tab on the ribbon.
- Click Freeze Panes.
- Select Freeze Top Row.

Freeze First Column

Use when the first column contains names or labels you need to see as you scroll across.

- Go to View > Freeze Panes.
- Select Freeze First Column.

ELECTRONIC PUBLICATIONS

DATABASE MANAGEMENT SYSTEMS

INTRODUCTION TO DATABASE

A database is a collection of related data , organized for easy access and retrieval when needed by the user.

A database is a structured collection of logically related data that is stored in a way that makes it easily accessible.

Databases can store data about anything e.g. about stock, people, sales, etc.

Examples of databases include Students database and Employees database

To work on a database there is a need for a database Management System (DBMS).

A DBMS is a computer program specially designed to create, organize and manage databases.

It is a software used for organizing and storage of data for easy analysis and retrieval.

Examples of DBMS

- Microsoft Access,
- Oracle, MySQL,
- dBase,
- SQL server,
- Oracle,
- MySQL,
- Informix,
- DB2,
- Paradox,
- FoxBASE,
- FileMaker Pro,
- Lotus Approach

Types of Databases

Databases can be manual or computerized;

Manual databases are referred to as paper databases which may include class registers, telephone books, dictionaries, business directories etc. A manual database is a hard copy file storage system that consists of paper records, folders and filing cabinets or storage boxes.

Electronic databases are also known as computerized databases are the computer-based databases and the common examples may include phone inquiry systems, voters" register, flight information, national teacher's registry, PUJAB admissions etc. An electronic database is a computer- based collection or listing of information saved in a way that makes it easily accessible.

Advantages of Electronic Databases

- Electronic databases store very large amounts of data
- Electronic databases allow easy input and editing of data
- Electronic databases enable automatic updating and re-calculating of data
- Electronic databases make it easier to query, search, filter and retrieve required data.
- Electronic databases format, arrange and present information in customizable ways
- Electronic databases can easily share the information with other software application/programs

- Electronic databases allow centralized use of information amongst many users over a network and therefore reduce duplication. e.g. in banks.
- Errors created during data entry are minimized since data is validated before it is entered.
- Many built-in functions are available to in Electronic databases which simplify calculations.

Disadvantages of Electronic Databases

- Electronic databases are expensive to set-up and manage.
- They require employment of specialist staff like database administrators.
- The failure of some components of the database may cause failure of a large number of operations. This is because very many users and applications rely on the DBMS.
- Complexity. Most database systems are hard to use and require special training of the staff.
- Electronic databases cannot be used without electricity.
- Electronic databases are vulnerable to virus attacks which leads to data loss.

Areas where a database can be used/applied

- Report card generation: a database can be used by schools to generate report cards and necessary academic summaries.
- POS (Point of Sale): in a supermarket, a database is used to design and automate a point-of-sale interface to manage money coming in, stock movement, e.t.c. E.g. Standard Supermarket in Kampala.
- Banks: a big database is used to manage details about a customer's transaction with the bank.
- Electoral commission: it manages a database archive for all eligible voters in a given country.
- Data warehouses: information bureau use a database to manage and distribute information to users for example information about air travel by various air companies.
- Stores: a database keeps consistent and reliable data. Very big stores used databases to store, manage and automate store records.

Common types of databases

A flat file database is made up of only one table. It is easy to setup and use

A relational database can take information from two or more tables, combines them into a new table or report using a key field.

Database Objects

These are the components that makeup a database. Examples of database objects include, tables, forms, queries, reports, modules and macros.

Table: A table is a primary database structure where data is stored. Tables store related data in rows (records) and columns (fields).

Form: A form is a database object that is used to enter or display data in a database. Any changes made to the data in the form is automatically reflected in the table

Query: is a database object used to request for specific type of data from a database table or combination of tables. Object used to search for or extract data from the table. A query picks information from the selected table (s) and then displays a subset of data that meets a given criterion. The process of asking a database for specific type of information is called **querying**

Report: A report is a database tool used for summarizing data in tables for easy reading and printing. Object that presents data from a table or query and places it in a neat, organized and readable way.

Working with Tables

A table is a primary object from which other objects are created. A table contains data about an entity.

An entity: is a person, place, thing, activity, or event for which data is collected

An entity is any object that has a distinct set of properties, for example, a student, a product etc.

Entity	Property
Student	Name, Age, Parent, address, Class etc.
Product	Name, Manufacturer, Expiry Date etc.

Commonly used data types in Microsoft Access

Datatype	Description
Short text(Text)	Stores names, words and numbers such as phone numbers and address. It is not used in calculations. Text or short text can accept a maximum of 255 characters
Number	Stores numbers that can be used in calculations e.g. Fees, Salary. Prices, Age, Marks etc
Date/Time	Stores dates and times in different formats e.g. short date, medium date etc.
Currency	Stores monetary values to 2 decimal places. And it prevents rounding off during calculations
AutoNumber	This is allocated by computer as number in sequence, i.e 1, 2, 3, etc. Each record is given a different number. This field is often used as primary key
Yes/No	Stores data such as Yes/No, True/False, On/Off, Present/Absent etc. the field must contain one of the two values given

Activity: Setting Appropriate Data types

ID NO	Name	Class	Date of Birth	Fees Paid (Shs)	Home Number
001	Patricia	S3A	7 June 2005	560,000	09876655443
002	Canny	S3D	5 Feb 2008	450,000	04567890987
003	Bonny	S3C	16 Nov 2006	480,000	03123456789
004	Faridah	S3B	23 Jan 2007	610,000	04123456900

Determine the appropriate data types for each of the fields in the table above

Key terms

A primary Key (PK) is a field that identifies each record in a table uniquely

Characteristics of a primary key

A primary key's main features are:

- It must contain a unique value for each row of data.
- It cannot contain null values.
- Primary keys are also immutable, meaning that once a primary key value has been assigned to a particular record, it cannot be changed or re-assigned.

A PK (Primary Key) ensures that no row have the same value. A table can have only one primary key and this key field must have values (Not Null). When you think about a PK field, think unique—each PK value must be the only one of its kind in a table.

A student ID and Invoice Number are two good examples of PK fields.

A field is a column showing a single piece of data about an entity such as Event title, Event start date. It is a column in a database table.

A Record is a row showing a collection of data about an individual item, such as an event, student, an employee, or a project. This is a row in a database table.

NB. Every table you create should have a primary key because it helps keep your data organized and easy to work with.

To “turn off” the PK, click the PK button. You will notice that the PK button is no longer “orange” and the little key is gone from the left of ID.

Table Field Properties

This is a specific characteristic of a particular field. Field properties are specified from the general tab of the database table structure.

Field properties include characteristics like:

- | | |
|------------------------|--------------------------|
| • Field size | • Indexed |
| • Caption | • Required |
| • Format | • Input mask |
| • Decimal place | • Validation rule |
| • Zero length | • Validation text |
| • Default value | |

Field Size: The value of this property determines the maximum number of characters that can be typed in a text field.

Format: This is used to set the layout of data in a field. For example, the Format **dd-mm-yyyy** sets dates to appear in the format 02-Jun-2014.

Input Masks: Input masks control how users enter data into a database. For example, an input mask like (000) 000-000000 can force users to enter telephone numbers in the Ugandan format i.e. (256)772-123456.

NB: A 0 in an input mask acts as a placeholder for a mandatory single digit. Users **must** enter a single digit in that position.

Validation Rule: Is used to check the value entered into a field as the user leaves the field. Only data that obeys this rule is a valid entry for this field. E.G The rule "F" Or "M" in the sex field accepts F or M and rejects anything else entered.

Validation Text: This is the text message that you wish to display to the user if the validation rule is violated.

Default Value: Specifies a String value that is automatically entered in a field when a new record is created. When users add a record to the table, they can either accept this default value or enter another value.

Required: You can use the Required property to specify whether a value is required in a field. If this property is set to Yes, you must enter a value in the field, the value cannot be Null or blank.

Example

Create a database and save it as Students

In the created database, create a table and save it as students table as shown below

Field Name	Data type	Description
StudentID	Auto Number	Format: Stacon-001
Name	Text	Maximum characters 25
DateOfBirth	Date/Time	Format: mm-dd-yyyy
Address	Text	
Contact	Text	Input mask: 0000-000-000
Gender	Text	
Fees	Currency	Format: UGX

Enter Any five records of your choice in the table above.

Table views

For tables, the common views are **two namely;**

Datasheet view: Enables the user to display and modify the table's data.

Design view: Enables the user to display and modify the table's structure. In the design view, you enter the field names and assign them data types.

Working with forms

A form is a database object that is used to enter or display data in a database.

These are tools that enable one to enter and display records from the database tables easily. They can be used to enter data in many tables at the same time.

Form Views

Form View – this view allows you to view, create and edit records

Layout View - this view is similar to Design View but is more visually-oriented in that each control displays real data. As a result, this is a very useful view for setting the size of controls, or performing many other tasks that affect the visual appearance and usability of the form.

Design View - this view gives you a more detailed view of the structure of the form. You can see the header, detail, and footer sections for the form. You cannot see the underlying data while you are making design changes.

Creating a Form

- Select the table or query on which you are basing the form
- Click on the Create tab
- Click on **Form** command in the Forms group

Working with Queries

A query in access is a way to retrieve data from one or multiple tables basing on a given criteria or condition.